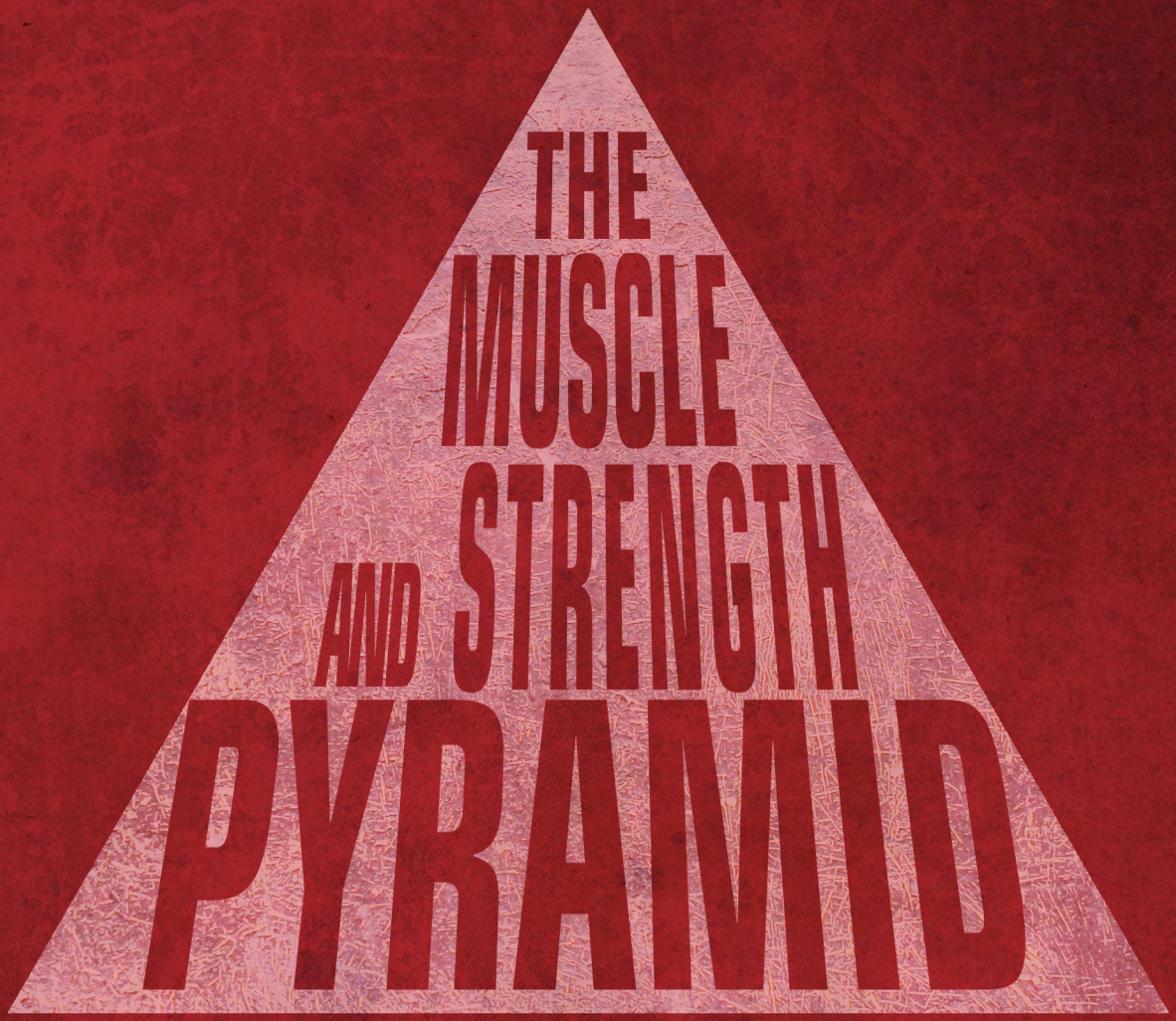

NUTRITION

ERIC HELMS
ANDREA VALDEZ & ANDY MORGAN

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Foreword

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I first gained awareness of Eric around 2005, when I became a moderator of the Bodybuilding.com forums. At the time, Eric already stood out as one of the most astute and helpful members in the online community. His passion for bodybuilding and fitness could be seen a mile away. I had a strong hunch that those who crossed his path would, at the very least, have a brush with greatness. As we fast-forward to 2015, it turns out my hunch was correct. A decade has slipped by, and Eric is not only my colleague, co-author, and fellow researcher/lecturer/practitioner, but he's also my close personal friend. He has amassed a wide range of academic and athletic accomplishments that would make anyone envious—or in my case, extremely proud and happy for him.

Watching Eric go from student to renowned athlete, coach, lecturer, and scientist has been very inspiring. Eric often credits me for being one of the key influencers in his career, but I must credit him for helping solidify my "status in the game," so-to-speak. When Eric approached me and our friend Peter Fitschen about collaborating on a colossal manifesto on bodybuilding nutrition and exercise, I was thrilled. When I found out that this project would be put through peer review, I was even more excited, since it was the only paper of its kind, and was sorely missing in the scientific literature. As I write this, "Evidence-based recommendations for natural bodybuilding contest preparation: nutrition and supplementation" is the third-most viewed article in the history of the prestigious Journal of the International Society of Sports Nutrition. With Eric at the helm of the project, I'm hardly surprised that it has ascended this far.

What you have in your hands is a book that every author in the nutrition & fitness world *wishes* he wrote. I secretly kick myself for not having conceived the presentation of nutrition (and training) elements in a hierarchy of importance in the utterly elegant way that Eric has done it. The organization of the book is logical and perfectly paced. It begins with the necessary scientific framework and foundations that no one should skip before proceeding toward the finer programming elements. Meticulous care has been taken to ensure that the reader is not left with blind spots or head-scratching moments. The book is rich with examples and real-world scenarios to help crystalize the concepts and processes.

Having been 'raised' in the formal academic curriculum, I can confidently say that if this book was a required text, we would have a new generation of nutrition students and teachers armed with all of the necessary tools and skills to help themselves and others succeed in this realm. The content is incredibly thorough, but also very focused. No fluff, no gimmicks, nothing of borderline relevance. A nice touch is its diligence in referencing the scientific literature, as opposed to the common practice of making bold claims out of imagination and thin air.

Whether you're beginning, intermediate, or advanced in your nutrition knowledge, you will benefit immeasurably from this book. Eric has taken all of his vast academic education and field experience with clients (and yes, himself), and pooled it together in a superbly written piece of work that will certainly become a classic. Enjoy every step of the journey!

– Alan Aragon

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PREFACE

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A few years ago I saw a need for a system that comprehensively covered the big picture of nutritional strategies, approaches, and theories for people interested in muscle and strength development. This decision was spurred by my years of experience working with individuals who wanted to improve their physiques through the manipulation of weight training and nutrition, but who all seemed to be lacking the same understanding of context and who were all plagued by black and white thinking.

I typically can tell if someone has some deep rooted misconceptions about nutrition based on the questions they ask me. Very commonly in the fitness field, you will be asked questions like "should I take the yolks out when I eat eggs?" or "Is bread bad?" Well, that's kind of missing the whole point. Does it fit with your energy balance? Does it fit in with the fat or carbohydrate intakes that you have available for the day? These queries sound perfectly reasonable, but in fact they imply a fundamental lack of awareness of what is important in the realm of nutrition.

As some of you may already know, I am one of the 5 coaches on a team called 3D Muscle Journey. We are a group of professional coaches who help people pursue their goals with drug-free bodybuilding, powerlifting, Olympic lifting, strongman, and anyone else who may be interested in strength or muscle mass gains. We do this in a couple of formats. We have our weekly coaching service which is primarily for people actually competing who need consistent ongoing guidance, and we also have our one-off consultations which can either be done as a standalone programming service or in sporadic sessions for off-season and general population coaching.

Without fail, something that I have learned from consulting with people in these one-on-one discussions is that typically, the reason they aren't getting to where they want to be is because they don't have a system. They don't have an understanding of prioritization, and they can't differentiate between big rocks and pebbles. I think that honestly, the fitness industry as a whole needs to take responsibility for this lack of awareness.

I can't tell you how many articles I run into with titles like "10 Good Foods that will get you ripped!" or "10 Bad Foods that bodybuilders should always avoid". These articles jump right into food choices without any mention of what your calorie intake should be, what your macronutrient composition should be, and pretty much just focus on the quality of foods when you look at them in isolation. The problem is that we don't eat food in isolation. We have diets. So as I like to say to my clients and to anyone who will listen, there's no such thing as a good or bad food, but there is such a thing as a good or bad diet.

It is because we in the industry present ideas and thoughts that are easy to turn into small sound bites that are "packagable" and sellable, but don't have any context, that people run into trouble. More often than not, the smart, well-intentioned, motivated people that I talk to have a lot of potential for success, however, they end up not making any progress because they have the proverbial cart before the horse.

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As you can see, we've got this fancy little cart here, it's got its lead, but it's hooked up to nothing and it's just hanging out here in front of the horse. Now the horse is sitting here going "Hey, I want to pull this cart, but I can't do it while I'm behind it." This expression of 'putting the cart before the horse' means that you are focusing on the details before you know the context of those details and the big picture.

For example, let's say you're an aspiring race car driver, and you've spent months researching race courses, strategies for competition, mechanical engineering as it relates to improving handling and top speed, and how to select a good pit crew. But you haven't yet learned to drive and don't have your driver's license. This may sound like a ridiculous error to make, but believe me, people do the equivalent of this every single day when it comes to nutrition.

So, I decided to design what I call "The Muscle and Strength Nutritional Pyramid". It is a five step pyramid with Level 1 as the foundation, the most important level. Then, as you move up into Levels 2 through 5 the level of importance gets progressively lower. I also acknowledge the overarching themes of mentality, adherence, lifestyle, and behavior that are linked to all levels of the pyramid. These concepts will be discussed in this text before, after, and throughout the explanation of the pyramid itself. While all elements of the pyramid are important, it's critical that we discuss the most important things before we discuss the *least* important things.

This is the whole reason I've made this pyramid. To help create some context and general guidelines for you to follow in an order that makes sense scientifically for reaching your goals.

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Lastly, I also want to let you know what this guide is not. In this book I take the position of focusing on what to do, not what not to do. Frankly, I get extremely tired of the constant stream of myth busting that goes on in our industry. The reactionary approach of "evidence based" professionals is just as maddening to me as the gurus who promote nonsense. Waiting around for a moron to say something stupid just to point out why it's stupid is a waste of time in my mind, and I'd much rather spend time helping you learn what to do, why, and how to do it. Got it? Good.

Let's dive in.

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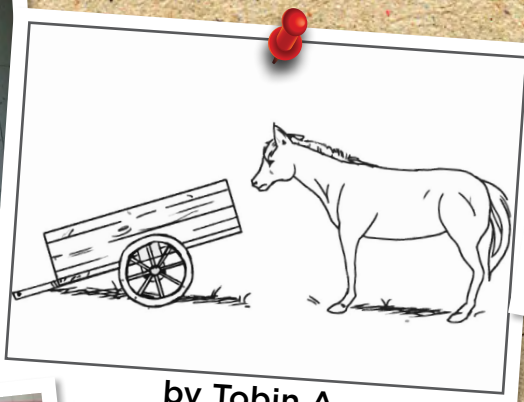

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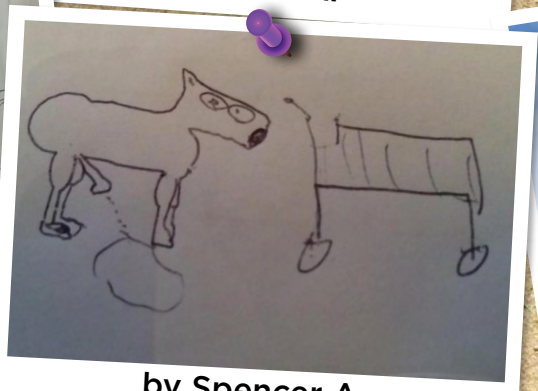

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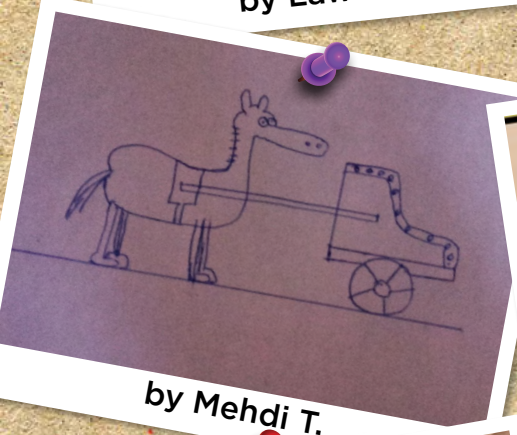

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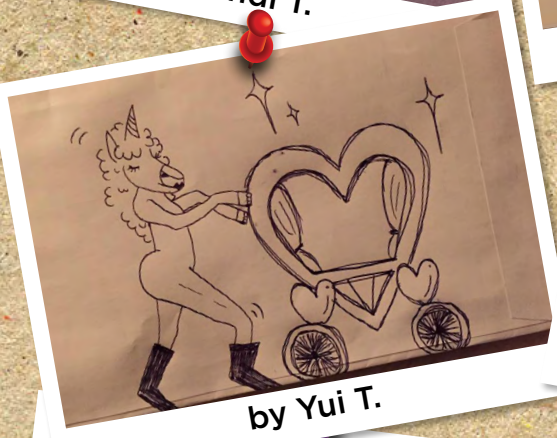

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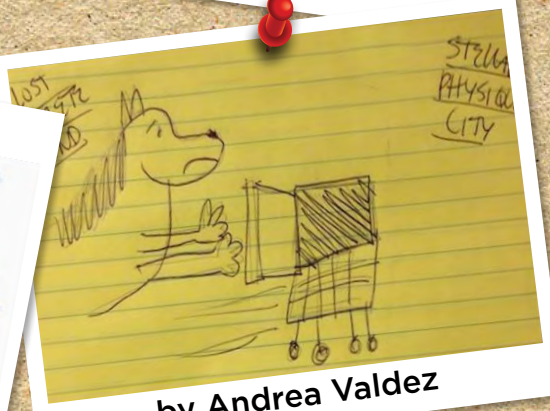

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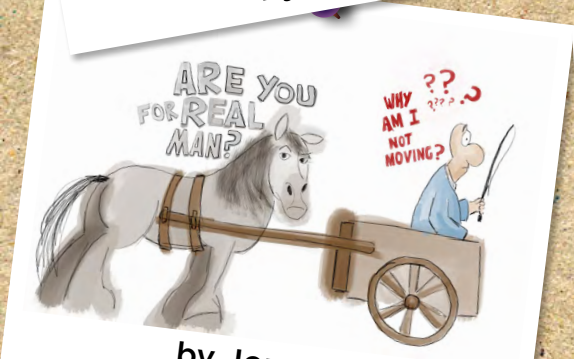

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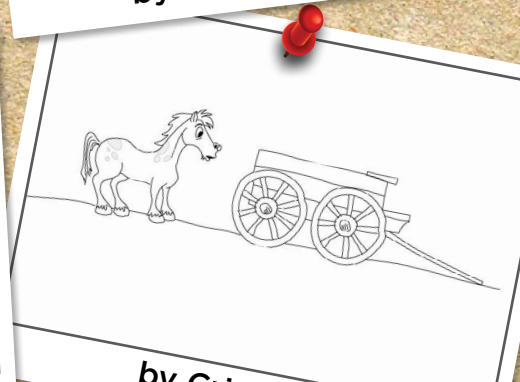

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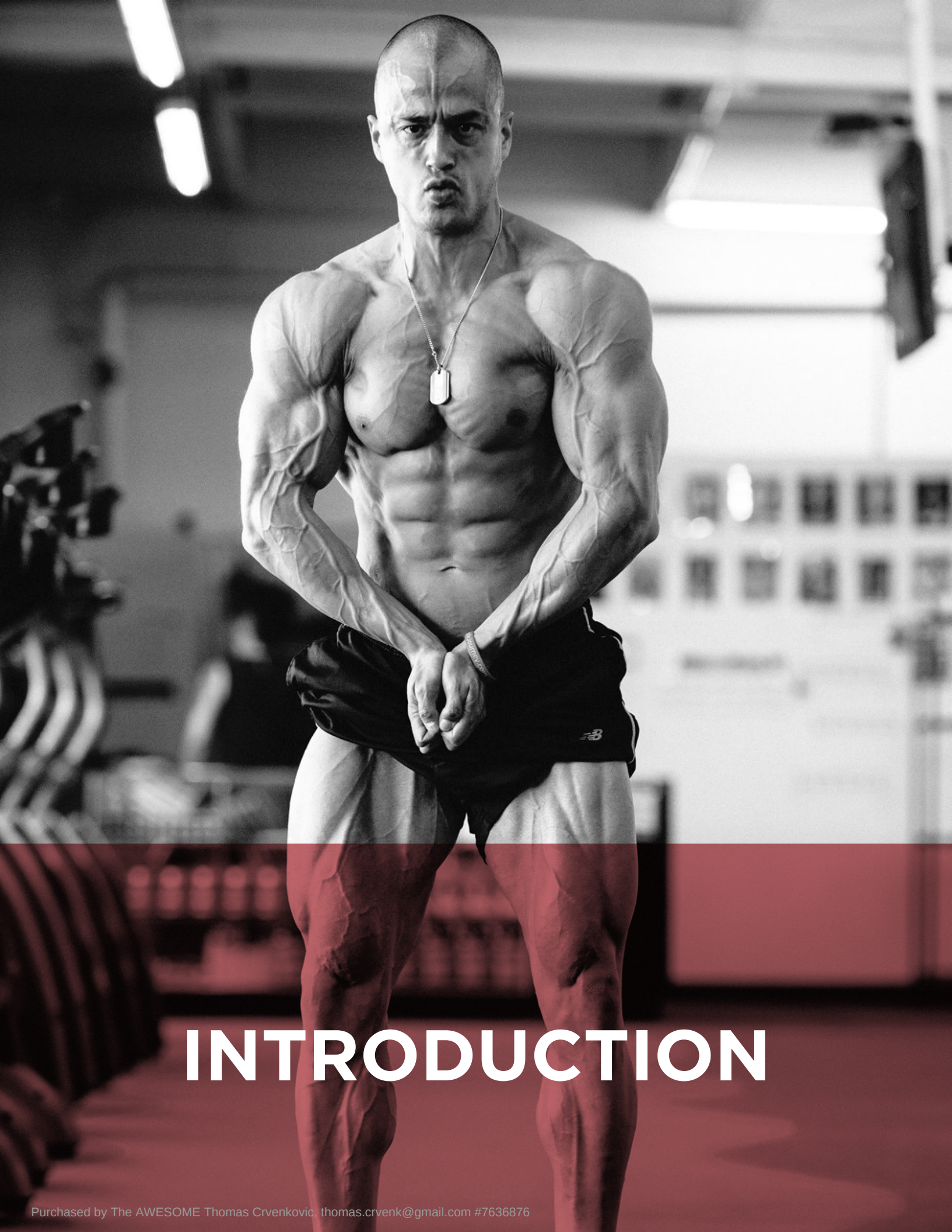

INTRODUCTION

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Let's start with a brief overview of what is included in the chapters of this guide.

Mindset & Materials

The beginning section will cover everything that is necessary to own and understand before implementing The Pyramid and its levels. There must be a balance between accuracy, flexibility, and consistency as we embark on our nutritional endeavors, otherwise adherence becomes a serious issue over time. We will discuss the psychological outlook required to keep yourself sane and satiated, and the physical tools needed to track your progress.

The Levels of the Pyramid

Once we have some basic understanding of what we will need upfront, we will then get into the meat of the matter. All levels of The Muscle and Strength Nutrition Pyramid will be very intricately described with many recommendations, values, and equations for setting up a sound nutrition plan as it pertains to your particular goals.

Now as a brief introduction to The Pyramid, here are the levels in order of decreasing priority.

Level 1: Energy Balance, Calorie Intake, & Rate Of Bodyweight Change

In this first and most important level of The Pyramid, we will go over how to determine your caloric needs, specifically as they relate to recommended rates of bodyweight change for muscle gain and fat loss.

Level 2 – Macronutrient Composition Of The Diet

Here we will go over the function of the macronutrients and then establish macronutrient intakes that are appropriate for most athletes in various conditions. I'll also introduce some alternative guidelines for those who may fall outside of the norm, and then discuss how to identify where you fall within the spectrum of all of these values. To close it out, we will go over fiber intake recommendations for health and nutrient absorption.

Level 3 – Micronutrients & Water Intake

This section will go over the types of micronutrients and how to use your food choices to fulfill their daily requirements. We will then finish Level 3 with fluid intake recommendations and how to tell if you are consuming enough to keep yourself hydrated and functioning properly.

Level 4 – Nutrient Timing & Frequency

This level discusses how to distribute calories over different periods of time. We will do this on both a broad and acute scale, diving into the timeline of the whole diet, over the course of a week, over the course of a day, and in relation to training.

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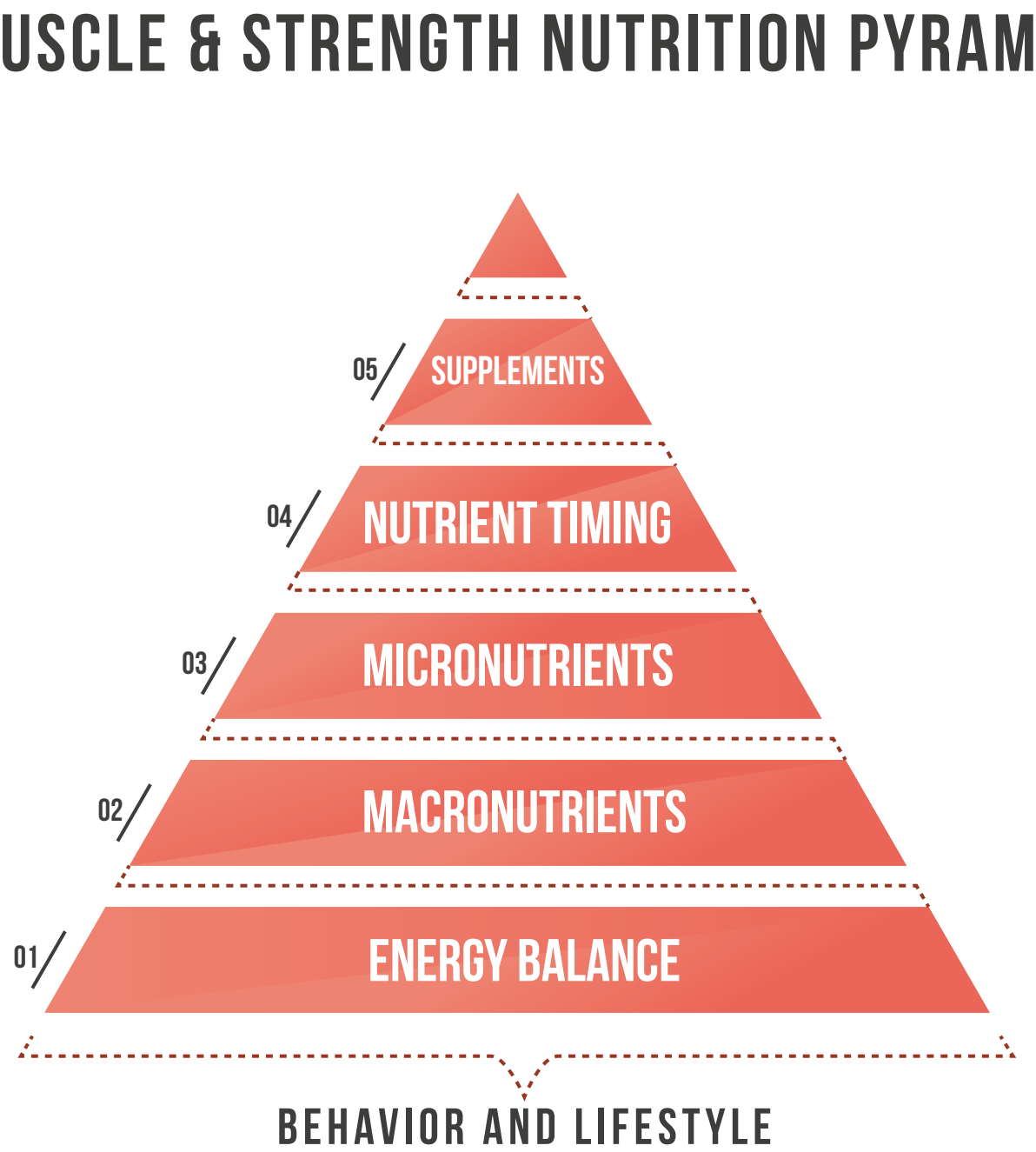
Level 5 – Supplements

Supplements are the least important part of The Pyramid and are not necessary for success, barring that you do not have some type of nutritional deficiency or condition that requires them. In fact, in the grand scheme of things, the vast majority of supplements on the market don't do anything except burn a hole in your wallet. That said, some are based on sound science and are therefore worth discussing as they can provide a small benefit. And because there are so many ineffective or unnecessary supplements on the market, we also need to discuss how to differentiate between the useful ones and the not so useful ones. Once we've learned how to evaluate them, I will present a list of recommendations and doses for you to consider.



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Behavior & Lifestyle

And lastly, in the final portion of this text we'll discuss how to implement these numbers and values into your current way of life. This can involve everything from tracking alcohol consumption, how to tackle eating out, and how to track in a less-strict manner as you become a more advanced dieter.

I will then close this guide with some of my ideas on how to better relate to other people in your life while you progress towards your goals. We will discuss the support of family and friends, and what to do when people ask you for advice.

As you can hopefully tell, The Muscle and Strength Nutrition Pyramid is not a rigid program to be followed in an extreme manner. I believe it is very valuable to create discussions around the implementation of your diet within your lifestyle throughout the process of changing your behaviors, so that is what you can expect throughout the remainder of this text.

To help you with the calculations in the coming sections, we have created a special page on the website for you with a spreadsheet calculation aid on it here. The password for the page is "nutrition-aid".

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MINDSET & MATERIALS

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We will start this guide by discussing your entire mentality towards nutrition, which can affect *all* levels of The Pyramid.

This is a broader and less definitive and quantitative topic than the rest of this book, but I can't express how important this is. You can have a lot of theory and a lot of knowledge and still never get to where you want to go if you don't know how to apply those concepts consistently and how to make them a part of your lifestyle and behavior.

This section will conclude with a discussion of the tools you need to implement the information in this guide and actually ensure that you're able to meet the guidelines of the levels within The Pyramid in accordance with your goals.

Accuracy, Flexibility, & Consistency

The implementation of The Pyramid is all about balancing these three qualities. If you become overly focused and accurate, you're going to lose some of your flexibility, you're going to lose your sanity, and inevitably you're going to make the diet very difficult to follow.

We only have so many things that we can focus on at a time, and our willpower and ability to juggle and handle multiple stresses is not infinite. You can't expect to chase down every grain of rice that falls off the food scale every time, 3 to 5 times a day, while hitting your macros with perfect accuracy and expect for that to be a sustainable plan. That will stress you out eventually or at the very least take your time and energy away from more important things in life.

What will stress you out even more than the process of trying to be overly detailed and accurate is what happens once you run out of energy to do this, and you can't do it consistently anymore. Then, you start to bounce back between the extremes of losing control completely and over eating, and rigidly tracking until you lose it again. Living in the two extremes is something to avoid, and to do so we really want to make sure that we have a balance of these three factors.

We want to be as accurate as we need to be in order to be consistent enough that we can get to our goals. Different goals are going to require different levels of accuracy, but they all require consistency, which means adapting your flexibility to your goal. In research, dietary restraint is highly associated with folks who can lose weight, but *flexible* dietary restraint is associated with those who lose weight, keep it off, and stay sane while doing so [1]. So determining the appropriate amount of flexibility for your situation is very important.

For example, a bodybuilder who is in the final stages of prep trying to get shredded glutes is going to need more accuracy than someone who has an undetermined time limit to lose 100 lbs. Likewise, a bodybuilder in the off-season trying to put on muscle mass is going to have different accuracy requirements from a bodybuilder during contest prep, or from someone who is trying to get lean but isn't a bodybuilder and is

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just trying to shape up for the summer.

The more accurate you are, perhaps the more consistent you will be in hitting your targets and achieving your goals. However, you might also be more stressed depending on how rigid you are in your pursuit of accuracy, which could negatively impact long-term consistency. The less accurate you are, the more flexible you might be. But if you are too inaccurate, and way too flexible to the point where you're not consistent, you won't get to your goals.

As you can see, this is a balancing act that must always be appreciated at any stage in your dieting process. It's important that we discuss different ways to align your dietary approach with your lifestyle and goals.

The Trouble With "All Or Nothing" Thinking

Now knowing that we have to consider the "accuracy to flexibility ratio" in order to improve consistency, let's go over one of the biggest mindset challenges that can get in your way of finding a healthy balance.

The approaches listed below can all end up being counterproductive in the long run because they fall under the umbrella of all or nothing thinking, which can be a huge short-circuit to many people's progress.

Rigid Meal Plans

Since I'm not a registered dietitian (RD), it's not within my scope of practice to provide specific meal plans. But even if I were an RD, I wouldn't write meal plans. The main reason I don't write meal plans for my clients, other than providing them as an example so my clients can initially see how to match foods to their nutrient targets, is because meal plans are often viewed in a binary fashion. Either one thing or the other, zero or one. People see it as, "I'm on the meal plan," or "I'm off the meal plan." In our minds, we then gauge success in black or white terms; I've either followed the diet and I was good, or I'm off the diet and I was bad.

For example, a meal plan might direct you to have an apple at noon, and instead you have a banana. If you view a meal plan as an immutable plan that must be followed, you may feel like you screwed up, get down on yourself, and because you see the day as already lost, eat a whole large pizza. The banana might have had the exact same macro-nutrition profile as that apple, but because of the mindset, you feel like you blew it and you go on a big binge.

I understand the allure of meal plans. They're easy and straightforward to follow. You don't have to learn about the macronutrient profiles of foods to follow them; you don't have to plan out your meals or your days on your own. It can be nice to just be told what to do. But you can't have a meal plan while you go on vacation, you can't have a meal plan and go out to eat, and you can't have a meal plan without getting into the mental traps of not knowing what to do when you're off it. They're not flexible, they don't teach

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you about nutrition, they tend to be socially isolating and when following them you don't learn how to integrate nutritional knowledge into your lifestyle. In many cases, meal plans won't help you get to your long-term goals. They are a short-term solution.

Consider that most people in this day and age are great at losing weight; the larger problem here is that they just can't keep it off. This is why we need long-term solutions, and this is why I don't advocate rigid meal plans.

That being said, there's nothing wrong with having meal plans so you can get an idea of how to hit your macros (which is a concept we will fully cover in Level 2 of The Pyramid). By seeing a couple of different examples, trying them out, and then learning how to diverge from the plans while still hitting your targets you can teach yourself to navigate your nutrition on your own.

So I would highly recommend that if you do use meal plans that they're a learning tool in your quest for flexibility and consistency, rather than a final outcome. Treat them like your nutritional training wheels.

One very useful website that develops meal plans for you is EatThisMuch.com. It is basically an algorithm where you plug in the calories you want, the number of meals you want to eat, the actual macros you want to hit, food preferences, and click "Generate". This website then spits out a meal plan for you relative to the targets you entered. It's a great way to add customized training wheels before you start free-styling with your macros.

"Magic" Macros

Once you've gone through the calculations in Level 2 of The Pyramid, depending on your goals, you may have a period of time where you are tracking certain macronutrient targets that you will aim for each day. This means you will have a particular amount of dietary fats, carbohydrates, and protein to consume each day, in grams.

That being said, they are still just numbers. You've decided they fit your needs based on this guide, but remember it's not like they are the end-all be-all. They will change over your lifetime as you get older, gain and lose weight, change your activity levels, or as you experiment and find out what works best for you.

To illustrate why being too attached is not necessarily a good thing, let's say you were supposed to hit between 190 and 210 grams of carbs. You "blew it" and ate 215, then because you have an all or nothing mindset, you go eat that same pizza we talked about earlier as if you'd been on a rigid meal plan. The problem here was obviously the pizza, not the 215 grams of carbs that put you 5 grams over your range.

So don't put your macros on a pedestal as this defeats the purpose of why we provide quantifiable numbers in the first place. You're not going to always be 100% accurate, sometimes you'll mis-track, sometimes you'll misread the label, sometimes you'll look up and just realize you just kind of blew it and went over. Just stop there, and get back

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on track the next day. Like a meal plan, the macros you are targeting aren't magic.

It's good to be consistent, but don't get into the black and white thinking of being either "on my macros," or "I'm off, so I might as well just give in and say screw it." That's what we want to avoid. The all or nothing thinking that eventually leads to binging. We want to majorly steer clear of that mentality, especially while we're dieting.

The "Good Food Vs. Bad Food" Approach

Now this is something that is very prevalent throughout the entire fitness community as a whole. Many people advocate it and have done so for years. These believers in the "good food vs. bad food" mindset include some very smart people, people with great physiques, and a lot of people that have had quite a lot of success with their own goals.

However, despite the fact that you can achieve success with this approach I don't see it as a long-term solution and I think that it can lead to developing unhealthy relationships with food. We shouldn't have to maintain borderline eating disorders just to keep a lean physique.

As opposed to trying to avoid the consumption of "bad foods", I think a better mindset to adopt is approaching nutrition with the goal of being inclusive instead of exclusive.

What this means is that we want to adopt an approach of *including* "healthy" foods that have a high micronutrient density and other desirable qualities (like a high fiber content), versus *excluding* foods that might not have these qualities. (*We will learn all about micronutrients later on in Level 3 of The Pyramid; for now, just know that these are the vitamins and minerals that your body requires.*)

There are very few, if any, foods that are actively unhealthy for you. Truly there are no foods that if eaten once, regardless of quantity, immediately and measurably harm your body. The only plausible negative connotation associated with say a Twinkie, a Pop Tart, and other foods commonly labeled as "bad" is that they are relatively devoid of micronutrients, fiber, and protein.

Some people refer to these foods as "empty calories", which is probably a slightly fairer description than simply labeling them as "bad". This term means that while these foods contribute to your calorie (Level 1 of The Pyramid) and macronutrient counts (Level 2 of The Pyramid), they won't do much to satisfy your micronutrient requirements (Level 3 of The Pyramid). While this description is relatively accurate, it doesn't mean these foods should be villainized and completely avoided. The main thing to be aware of is that "empty calorie" foods can only cause issues if they completely dominate your diet. It's not that we need to remove them entirely; it's that we need to make sure that we have *included* the "healthy foods" first to ensure our bodies are nourished and taken care of. After that, feel free to have the "bad foods" (which really aren't bad at all) in moderation as this will improve your flexibility and therefore your consistency. By allowing yourself to diet while consuming a wider range of foods that might include "treats" in moderation, you will feel more normal, have more flexibility, less restriction,

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and ultimately more long-term adherence and success.

This is the reason that the seemingly normal approach of eating "good vs. bad food" or "clean vs. dirty food" can potentially cause problems. While it is true that high-level bodybuilders have and will continue to be very successful eating only from a short list of foods that are deemed by the bodybuilding Gods to be "clean", it doesn't mean that this is the only approach that can deliver success (it also doesn't mean that they can adhere to this approach after their diet has concluded).

Remember that you don't get extra credit for eating *only* healthy foods. Once you've met your basic requirements you don't get gold stars for consuming additional micronutrients. There's no food critic in your throat who tells you "this is good, this is bad, this is good, etc.," there's just your body getting its nutrient needs, and once it gets more than enough, it doesn't continue to benefit from more. It's not a question of whether a bowl of oatmeal is better than a candy bar. Rather than assessing which food is good or bad, you need to assess if your entire diet is good or bad. Believe it or not, a rigid "clean vs. dirty" diet can actually result in a poorer nutrient profile than an approach that includes a broader spectrum of foods.

One of the most critical components to a healthy lifestyle and diet is nutritional variety. For example Okinawans (the people who inhabit the island of Okinawa, Japan) are some of the longest living people on the planet. They also happen to have one of the highest food varieties of any culture in the world [2]. In contrast, there are some bodybuilders who want to be this healthy, but simply aren't because of the rigidity of their diets. Claiming to be all about health, but only eating from a list of 10 foods is an ironic proposition. If you have to cut out gluten, dairy, red meat, "processed" foods, fruits, legumes, starches and whole eggs, you end up with an incredibly limited diet that is unlikely to satisfy your micronutrient needs [3]. I've seen it time and time again (and experienced it myself) where following a very rigid diet results in the self-fulfilling prophesy of eventually being unable to digest foods not on the "clean list" without incredible discomfort due to the loss of the enzymes and gut bacteria that are essential to the digestion of a wide variety of foods.

Tools For Tracking

If you are a competitive bodybuilder going through contest prep, or a powerlifter dropping a weight class, having a high degree of accuracy and consistency is paramount. To do this, requires quantifiable ways of ensuring you are meeting your intake goals and making progress. This requirement is fulfilled by tracking bodyweight and aspects of your food intake.

For people who aren't competitive athletes, there is also value in tracking at certain stages of your journey. When you are starting out, tracking helps you understand the effect that your eating habits have on your body and tracking also gives you feedback on how successful you are at modifying those habits. Additionally, there are time points

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throughout your life where it is useful to once again track your food and weight. These time points occur intermittently such as when you go through dieting phases, as a "check in" to evaluate why progress might have ceased, or as a "refresher" to ensure the habits you've picked up are still serving you the way you think they are (more on this to come at the end of this book). The two things you need to track your intake and your progress are a food scale and a bodyweight scale.

I would highly recommend that both of these items be digital, as they are much more accurate. You can find them from $10 to $30 in most stores in the U.S. or online, and all you have to do is just make sure that the batteries don't run out. You don't want one of those non-digital scales that are loaded with a little rubber band because they are very inaccurate.

How To Track Food

The reason that you should weigh foods, as opposed to measuring their volume (when possible), is because measurements of volume are often inconsistent in comparison to weight.

One cup of the same food can produce very different values depending on the form of the food, how you pack the food into the cup, and whether the cup is level or not. A cup of diced apples, apple slices and a cup of mashed apple have different nutritional profiles. Also, cups are not uniformly manufactured. For example a food label might say "1/2 cup oats (45 g)", but when you actually use the cup you have at home and weigh the oats out, the scale reads 54 g, or 40 g. When you look at a food label and it states "100 grams (1 cup)", be mindful that 1 cup of this food (using the measuring cup you have at home) may not be 100 g. To ensure accuracy, weigh foods when possible.

Additionally, it's a good idea to weigh foods in their uncooked state. How long a food is cooked for affects how much moisture it retains. A food with higher water content before cooking weighs more, but it has the same nutrition profile after it is cooked when it weighs less after the water has evaporated from cooking. Since you won't always cook foods for the exact same length of time, a good way to ensure consistent measurement and tracking is to weigh foods prior to cooking.

You will also want some way of tracking the nutrients in your foods. This can be done on a digital spreadsheet, with a piece of paper and a pen, or using an integrated online food database (which is probably the easiest and most convenient). Some popular databases are MyFitnessPal, LiveStrong, FitDay, MyMacros, and CalorieKing, but there are many others. These computer-based applications can also usually be accessed on a smart phone which is useful while traveling, going out to eat, going to a grocery store, or eating anywhere else outside of your normal element. Just be aware that there can be mistakes in these databases, due their reliance on user input. So anything you consume often is worth checking in multiple places.

As we make our way through The Pyramid we'll discuss specifically which nutrients to

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track and what target ranges are appropriate for you given your goals and profile.

How To Track Bodyweight

Bodyweight is one of the most important variables we can track to gauge if we're actually getting to our goal.

In the first level of The Pyramid we'll discuss the appropriate rate of bodyweight change for your goal. After going through Level 1, you may determine that your current goal is to lose 0.5 to 1% of your bodyweight per week, or to gain 2lbs (~1 kg) per month, which means you're only putting on ~0.4-0.6 lbs (~0.2-0.3 kg) per week. But, how do you measure something as small as a 0.4 lbs (0.2 kg) increase in a weekly time period if you only weigh in on Saturdays under very different conditions? You can't.

What I recommend is to have a daily weigh-in first thing in the morning, after you use the bathroom, before you eat or drink anything, nude, and record the number. It's not that we care about the single day's weigh-in (we don't, and focusing too much on your daily weight can drive you crazy), rather it's that you're going to use your daily weigh-ins to generate an average weight for the week.

This can be easily automated using an Excel (PC) or Numbers (Mac) spreadsheet. Both programs have an automatic "Average" function built into their software. There is also a phone app called Happy Scale that averages and smooths out your weigh-ins for you to predict actual weight loss over time. You could also just do it long hand with pen and paper. Or you can search the web for "average calculator" and Google will have quite a few options for you to use for free as well. There are many ways to do it, but the main output we need is a 7-day average, compiled from weigh ins at roughly the same time of day, under the same conditions, to get as consistent data as possible. Once you have your average, the goal is to compare your average weight from one week to the next. Over a week, an average will flatten out daily fluctuations in bodyweight and give you a workable, reliable number.

It is totally normal for your bodyweight to fluctuate by 1-2% on a daily basis due to shifts in water (sometimes more for some people). This is caused by day-to–day fluctuations in food intake, sodium intake, alcohol, and stress hormones or from hormonal shifts during certain phases of a menstrual cycle (among other things). But what you'll notice is that when you get a weekly average, that number is much less variable and much more comparable when looking at a previous week's 7 day average. These averages become even more consistent if you perform the weigh ins as I recommend (first thing in the morning, nude, after you use the bathroom, before you eat or drink anything). Additionally, the weigh ins will be more reliable when your diet is more consistent (which this guide will help you do), and even if you do find that there is some variation comparing one week to the next, you will be able to see those trends and you can even decide to track 14 day averages if needed to make sure that you can tell if you fall within the coming guidelines for increasing or decreasing your body weight.

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Level 1 ENERGY BALANCE

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And now we dive into the actual hierarchy of The Muscle and Strength Nutrition Pyramid. As mentioned in the Introduction, these concepts will be delivered in order of importance from the foundation of The Pyramid (Level 1), to the top of The Pyramid (Level 5).

Each level provides the foundational knowledge for subsequent levels. Too often, information in Level 4 or 5 is delivered without context in the fitness world and this creates a lot of confusion. For example, a recommendation about when to consume protein is fine and good, but only if you understand what protein is and how it fits in with your energy and protein needs. The foundation levels of The Pyramid are the levels that give you the biggest return on your intellectual investment, and often covering these bases makes it much easier to understand and implement many of the concepts that are covered in the higher levels.

This is why the system is to be followed as such. In order, one step at a time, and always with respect to the preceding Levels.

That said, for most people looking to benefit from the information in this guide, their goals fall into a couple of different categories. There are those who are trying to acutely put on muscle, and those who are trying to lose body fat.

Now even though most people typically want to achieve both of these outcomes, trying to do both simultaneously, while certainly possible, is not necessarily the most efficient approach to achieving these goals. Certainly someone who is a rank beginner (or returning to training after a period of detraining) may be able to make progress in both areas at the same time due to having a low threshold for adaptation. But after a certain point, it may make more sense to focus on one or the other at any given period of time. So if you're a late-stage novice, intermediate, or advanced trainee, your efforts would best be aimed toward either fat loss or muscle gain.

In this first and most important level of The Pyramid, we will go over how to determine your caloric needs, specifically as they relate to recommended rates of bodyweight change for muscle gain and fat loss.

Maintenance Calories

To determine the appropriate amount of calories to consume for our planned gaining or cutting phases, we must first establish a baseline target that would theoretically result in no change. The estimation of caloric intake that would simply keep you at your current body weight is referred to as your maintenance calories. To determine maintenance, you can either track bodyweight and calories to ascertain the relationship between the two variables, or you can estimate maintenance calories with an equation. Both ways give you a value for maintenance calories, but the former is ideal for reasons I'll explain.

Finding Maintenance By Tracking Weight & Food Intake Over Two Weeks - My Preferred Method

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The ideal method that I'll introduce first requires more time, energy, and thought, but results in a much more accurate, individualized estimation of maintenance calories. To begin, you must obtain a digital bodyweight scale, a digital food scale, and an online account with a free food tracking database like MyFitnessPal (or others). Earlier in this book we covered specific recommendations on how to weigh, track, and measure. So before you decide to go through this process, make sure to look back on the Mindset and Materials section of this guide.

Once you've gathered these resources, the next step is to spend 2 weeks weighing and tracking your food as best as you can when you prepare it, and estimating your caloric intake when you aren't preparing your own food as accurately as possible. Once again, earlier in this guide we covered how to be accurate when not preparing your own food, so go back and read this section as I want to re-emphasize that you need to be familiar with tracking and measuring before starting this process. The goal of food tracking is to get a daily caloric intake that closely represents what you eat on a day to day basis. In conjunction with tracking calories, you will also step on the scale every morning, in the same conditions (like we covered in the previous section of this book) and record your bodyweight.

At the end of two weeks, you will have 14 weigh-ins, and 14 calorie values. First determine your average caloric intake by adding all 14 calorie values together and dividing by 14. Let's say as an example, this value came out to 2100 calories.

Then, split your weigh-ins into 2 weekly averages, by adding your first 7 weigh-ins together and dividing by 7 and doing the same thing with your last 7 weigh-ins as seen below.

Bodyweight	
Week 1	Week 2
141.7	141.8
142.1	142.2
142.7	143.0
141.7	141.7
142.5	142.5
141.9	142.8
142.1	142.0
Average ~142.1 lbs	Average ~142.3 lbs

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Now you see that you are slowly gaining weight. But what does this mean with regards to calories? Well, we know that 1 lb (~0.5 kg) of adipose tissue contains ~3500 calories [1]. Therefore we can estimate how much of a surplus or deficit you are in based on weight change. While "the 3500 calorie rule" isn't perfect since not all weight gained or lost is fat and energy expenditure changes over time [2], this value is still a good way to roughly estimate maintenance calories.

Reason being, you are using a short-term period of 2 weeks and you haven't manipulated your diet to result in significant changes in bodyweight or expenditure yet. Thus, changes in weight are predominantly fat and water and using a 7-day average will eliminate the "noise" of water weight fluctuations. So, you can be reasonably sure that the short-term changes in bodyweight you observe will be well represented by the "3500 calorie rule".

The only time I wouldn't advise using this method is if you just started training or are just coming off from a training layoff or injury, as in this case muscle mass can rapidly change and potentially invalidate the equations. However, if you are a natural lifter not coming back from a layoff, not a complete novice lifting for the first time, and if you are relatively weight stable, this is a very easy way to estimate maintenance calories with appropriate accuracy.

Hypothetically, if you gained 1 lb (~0.5 kg) of fat in a week, you would be, on average, consuming ~500 calories more than your maintenance per day, as 7 days x 500 calories = 3500. So if you only gained 0.2 lbs (~0.1 g), you simply multiply 3500 x 0.2, which is 700 calories. Meaning, in the above example you are in a weekly surplus of ~700 calories.

On a day to day basis, on average, that means you are consuming 100 calories more than your maintenance (700 calories divided by 7 days). Thus, you know that since your average intake over 14 days was 2100 calories, you know maintenance is roughly 2000 (2100 – 100 calories).

That said, if there is some sort of time constraint keeping you from this 2-week recommended method of determining maintenance calories, a secondary approach would be to roughly estimate using some basic calculations. Just be warned that it can be far less accurate than actual tracking, but can still yield some sensible results.

Estimating Maintenance By Calculation

Now keeping in mind that there are huge amounts of individual variation here, a good way to ballpark this for most people would be to first take your bodyweight in pounds and multiply it by 10 (or multiply your bodyweight in kilograms by 22) and then multiply that value by an activity multiplier. With me as an example male that weighs 200 lbs (90 kg), my theoretical baseline is 2000 calories. (I'm actually a little over that bodyweight, but we'll make it a nice even number for demonstrative purposes.)

Step 1: Baseline Multiplier; Not Accounting For Any Activity

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200 lbs x 10 = 2000 calories

Then we multiply that by an activity multiplier. Now when I say "activity", it's not just dependent on whether you're a construction worker, or sitting at a desk all day, a full-time athlete, or a weekend warrior. This also includes something called NEAT, which stands for non-exercise activity thermogenesis. Or more simply put, any activity outside of exercise, including subconscious movement.

So yes, someone sedentary will have a lower maintenance than someone highly active, but a part of the reason there is such large individual variation is that NEAT is one of the main ways we regulate energy expenditure. Meaning, if we diet and lose weight, or eat more food and gain weight, our body typically will adapt to some degree to maintain our "normal weight". It does this by making us burn more or less calories, and a lot of that is a function of just fidgeting, postural support and control, and subconscious movement. It's not something we can really modify. Some people will sit at their desk all day and burn a whole lot more calories than I ever will, simply because I have a "thrifty metabolism" [3].

Additionally, almost everyone reading this book will probably (or should, in most cases) fall in the range of lifting weights 3-6 times per week. So, taking into account individual variability and also your exercise schedule, we can use the numerical range of 1.3 to 2.2 as an activity multiplier. From my experience, I would say about 90% of people fall in this range.

So what that means is that yes, there are going to be people who need 2600 calories to maintain 200 lbs (not very much for a person that size), all the way up to maybe 4400 calories (a lot for a person that size). Most will fall in the middle range, but some won't. So there really is a large variation, and there are people who are even further outside of those ranges believe it or not. Below, I show how to use the activity multiplier, notice there is a range at each level to represent individual differences.

Step 2: Using An Activity Multiplier

LIFESTYLE & TRAINING FREQUENCY	ACTIVITY MULTIPLIER
Sedentary plus 3-6 days of weight lifting	1.3 – 1.6
Lightly active plus 3-6 days of weight lifting	1.5 – 1.8
Active plus 3-6 days of weight lifting	1.7 – 2.0
Very active plus 3-6 days of weight lifting	1.9 – 2.2

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Baseline Multiplier x Activity Multiplier = Estimated Calories For Bodyweight Maintenance

Depending on lifestyle and individual differences, these calculations can equate to a 2600 - 4400 calorie range for maintenance in our 200 lbs (90 kg) male example.

Now that we know how large of a range we're looking at mathematically, it is easy to see why it is best to actually take the two weeks to fully track our intake as discussed earlier. However, going somewhere in the middle here would yield a decent guess for estimating an appropriate daily caloric goal.

Rate Of Weight Loss

The next step in setting up a diet is to take maintenance calories and use that information to determine an appropriate caloric intake for our goals. For weight loss, I'd recommend that we aim to lose weight at a rate of 0.5 to 1.0% of body weight per week to minimize muscle and strength loss [4]. With the 200 lb (90 kg) male example we talked about earlier, this would be 1 to 2 lbs (~0.5 to 0.9 kg) per week.

As previously discussed, 3500 calories roughly equates to 1 lb (~0.5 kg) of adipose tissue; so if you drop 500 calories per day, then you'll drop roughly 1 lb (~0.5 kg) per week if the majority is fat. Again, this is not an exact science, but it's a good rough estimate to start from.

So, if we know our 200 lb (90 kg) male's maintenance calories, we could subtract 500 to 1000 calories per day from that total to elicit the appropriate rate of weight loss of 1 to 2 lbs (~0.5 to 0.9 kg) per week.

Now although this losing and gaining thing seems like simple math (i.e., "if I go down 500 calories I drop a pound, so if I eat 500 calories I gain a pound"), it often doesn't work out that way. Remember that whole subconscious activity and metabolic regulation thing that we were talking about? That concept means that you may have to eat more or less than you expect to get the rate of weight loss or gain you desire.

Additionally, we don't only lose body fat during a diet. In fact, muscle loss and potentially muscle gain while dieting can confound the numbers (more on this to come). Point being, these recommendations are just good starting places to set up the diet, but from there you will have to adjust.

Cardio For Fat Loss

Also, the caloric deficit doesn't have to come entirely from the diet, and you probably guessed that adding some cardiovascular work to expend more energy rather than restricting your energy intake alone could also be useful.

You burn approximately ~0.2, ~0.45 and ~0.7 kcal per 10 minutes per pound of bodyweight doing light (RPE 2 to 4 out of 10), moderate (RPE 5 to 7 out of 10) and vigorous (RPE 8 to 10 out of 10) cardio respectively, above what you would normally be burning

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doing everyday light activity in that same time period [5]. Cardio mode, height and other variables affect these values, but these are decent values to use for estimation purposes. So for example, a 200 lb male performing moderate intensity cardio would burn an additional 90 kcal in 10 minutes over and above what they burn doing normal day to day light activity for the same time period. In an hour, they would burn 540 kcal over what they would have burned had they been performing light every day activity.

Below is a chart displaying the amount of calories burned during 10 minutes of cardio activity at 3 different levels of intensity for individuals at 3 different bodyweights:

Type of Cardio	120 lbs (54 kg)	160 lbs (73 kg)	200 lbs (90 kg)
Light	24 kcals/10min	32 kcals/10min	40 kcals/10min
Moderate	54 kcals/10min	74 kcals/10min	90 kcals/10min
Vigorous	84 kcals/10min	112 kcals/10min	140 kcals/10min

Now if this hypothetical 200 lb (90 kg) male really enjoyed food he might think, "Hold on, if I did an hour of moderate intensity cardio a day, that would put me slightly over a 3500 kcal deficit per week and I would be able to lose a pound weekly which is at a rate of ~0.5%. That's what you prescribe and I wouldn't have to restrict my food!" Well, he wouldn't be wrong, but 7 hours of moderate-intensity cardio per week can cause problems for someone interested in muscle and strength.

Doing cardiovascular exercise at moderate intensities is essentially endurance training. The adaptations and the work required to produce endurance adaptations can interfere with the training and adaptations required to generate muscular strength, hypertrophy, and power [6]. Not to say that interference will *prevent* someone from getting bigger, stronger, or more powerful, but if excessive cardio is performed it can *slow down* the process of building muscle, strength, or power in a dose-dependent manner.

Glycogen depletion and the molecular signaling that comes from endurance training may play a role in interference [7]. Additionally, interference might also be related to the extent of the impact and the contribution of eccentric actions from the modality of cardio, considering that cycling appears to interfere less with resistance training adaptations than incline walking [8]. Eccentric actions are essentially when your muscle lengthens while it contracts, often performed when guiding a load into place or decelerating a load; like what your bicep is doing when you set down a coffee mug. In endurance training this is how your body brakes and controls your inertia and movement. High impact forces can create joint strain, and a high volume of high force eccentric actions

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can create a lot of muscle soreness. So, you can deplete the muscle of its energy and also go into training with sore joints and muscles if cardio training is excessive.

However, low-intensity cardio (if it doesn't have an impact component, like cycling, or the elliptical) would be below the threshold of producing overload and therefore wouldn't be an issue. For someone in decent shape who is lifting weights, casual easy cardio is not an adaptive stress, so it won't cause endurance adaptations in the body. Thus, interference is not an issue with low-intensity cardio. However, the calorie burn is much lower when doing low-intensity cardio compared to higher intensities and thus you have to do a lot of it for it to add up.

The final option is high-intensity cardio. High-intensity cardio is very taxing, and unlike lower-intensity cardio cannot be done continuously for very long unless you are a well-trained anaerobic athlete. This is one of the reasons you often hear people talk about high-intensity interval training or HIIT. HIIT is when you do a burst of maximal-intensity cardio, followed by a rest period and then repeat. Similar metabolic adaptations can come from either HIIT or lower-intensity cardio performed for longer periods, but in less total time [9]. Also, the risk of interference seems to be reduced when HIIT is utilized because the high intensity nature of the cardio is more similar to resistance training [10]. Lastly, the higher the intensity, the greater the increase in metabolic rate in the short term period afterwards. High-intensity exercise provides a short term, small, but significant increase in metabolic rate [11] while low intensity does not.

So does that mean HIIT is a home run and that our 200 lb (90 kg) male who loves to eat can do a bunch of HIIT and some low-intensity cardio and keep all his food? Well, unfortunately some of the same issues that come with moderate-intensity cardio come with high-intensity cardio. If there is a significant eccentric component, or a high level of impact, it can cause problems. In fact, sprinters suffer more than twice the number of hamstring injuries that long distance runners suffer on average, despite running only a fraction of the distance or time [12]. While a larger risk of injury (with certain modalities) and a greater need for recovery are the only risks of HIIT, these are significant risks. It's hard to make the argument that you are avoiding interference and retaining more muscle by avoiding moderate-intensity cardio and doing high-intensity cardio when you have a hamstring tear.

Okay, so then what is appropriate for cardio prescription?

Because of interference, cardio should not be the primary vehicle for fat loss, regardless of whether you perform low or high-intensity cardio. The majority of fat loss should come from the diet. Secondly, resistance training performance is the most critical aspect to muscle maintenance. The diet supports the training as best as possible while creating fat loss and the training supports muscle retention. Don't put this paradigm at risk. Remember you are a strength athlete or a bodybuilder, not an endurance athlete. Your total cardio for the week should take no more than half the time you spend lifting weights. So if you spend 90 minutes 4 times per week lifting weights (6 hours), that

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means you should do no more than 3 total hours of cardio per week. As an aside, smaller women may find that they reach a point where food can not be realistically further reduced to continue losing weight; in these cases it may sometimes be needed to max out the amount of cardio performed or even to go slightly above this amount.

Choose cardio that is easy on the joints (low impact) and easy on the muscles in subsequent days (won't make you sore). Rowing, cycling, swimming, elliptical trainers or even lightweight barbell or kettlebell complexes could all be used. Cap the number of HIIT sessions at one to two sessions per week that last no more than 30 minutes. Do no more than an hour per week in total of moderate-intensity cardio as this intensity causes the most interference. For the rest of your cardio, keep it at a low intensity.

So what might this look like?

In the example of lifting 6 hours per week, you could perform two 30 minute HIIT sessions, 1 hour of moderate-intensity cardio, and 1 hour of low-intensity cardio as one way of doing the absolute maximum amount that should be performed.

Rate Of Weight Gain

For someone who is aiming to put on muscle mass and strength, I'm going to give you some different figures depending on your training experience. Realistic and effective rates of weight loss are very different from weight gain; anyone who is a natural lifter and has been at it for 3 to 5 years will tell you that watching muscle grow is like watching paint dry, while fat loss can happen relatively quickly in comparison.

Even for a lifter who has been in the gym for a year and a half, you can massively change your physique in a 24-week contest prep or cutting phase by dropping body fat. However, it might take 24 months to make a massive visual difference in a physique on the same level during a gaining phase.

Now I'd like to clarify, when we discuss gaining muscle throughout this guide, I don't recommend just eating everything in sight and gaining massive amounts of weight through a classic "bulk" phase as previously identified in the old school bodybuilding community. That continual addition of body fat will eventually result in a lower strength to bodyweight ratio for someone who would like to get stronger, and it will make dieting much more difficult and prolonged for someone who would like to eventually cut down for the bodybuilding stage. In fact, a recent 12-week study comparing rates of weight gain in resistance-trained athletes found that the group that consumed a small surplus gained the same amount of muscle and strength, but only one fifth the body fat as the group that consumed an additional 600 calories [13].

Now that doesn't mean there is never a time to gain relatively quickly, it just depends on how much "gaining capacity" you have. This is determined by how long you've been lifting weights and how much of your muscular potential you've realized up to this current point. So to establish rates of weight gain, we will differentiate between

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beginner, intermediate, and advanced lifters with gaining goals.

Below is a chart referring to the recommended rate of weight gain per month:

Experience Level:	Definition	Rate of Monthly Weight Gain
Beginner	Able to progress most training loads in the gym on a week to week basis	1 to 1.5% of bodyweight
Intermediate	Able to progress most training loads in the gym on a month to month basis	0.5 to 1% of bodyweight
Advanced	Progress is evident only when viewed over multiple months or a year	Up to 0.5% of bodyweight

Now remember that if you go faster than this, the weight you are putting on will probably be proportionately more body fat. It is very easy for fat mass to come right along with muscle gain if you are not careful, which is something we do not want.

For a beginner example, a male who weighs ~180 lbs (~82 kg) would be looking to gain ~2-3 lbs (~0.9-1.3 kg) per month, while a female weighing ~130 lbs (~59 kg) would be looking to gain ~1-2 lbs (~0.4-0.9 kg) per month.

For intermediates, this will slow down a little bit because you are closer to your genetic potential of muscle mass, and now we're talking somewhere between 1-2 lbs (~0.4-0.9 kg) a month for our 180 lb (82 kg) male example, or ~.5-1 lb (~0.2-0.4 kg) a month for our 130 lb (59 kg) female example.

At the advanced level, we're primarily going to be looking for progressive overload and recovery in the gym. A very slight surplus is all that is needed, and at most a rate of weight gain of 0.5% of bodyweight per month should be targeted, with the understanding that this may still require periodic mini-cuts to keep body fat in check.

So why aren't we primarily focused on seeing a change in weight at the advanced level? If you look at any top-level professional natural bodybuilder, maybe Brian Whitacre, Alberto Nunez, Jeff Alberts, Patricia Beckman or any one of these competitors in the advanced levels of their careers, their stage weight might only go up or down 1 or 2 lbs (~0.5-0.9 kg) from competition to competition.

Yes, bodyweight increases will come over time, but it's going to be the amount that an intermediate trainee gains in 1 to 2 months over the course of 3 to 4 years. This is why looking at it on a monthly basis just doesn't make sense for the advanced lifter. If substantial increases in stage weight do occur in these athletes, typically it's a case of better muscle maintenance while dieting versus muscle gain in the offseason. So at this level what you want to focus on is making sure that you're recovering from your training and that you're making progressive increases in the gym. This is indicated by either adding more reps, or more load over time.

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These advanced individuals also know that they're past those introductory phases where strength improvements are occurring quickly, primarily due to neuromuscular adaptations or making improvements in form. Advanced lifters can be relatively certain that increases in reps or load are at least partially because they are making some subtle changes structurally. Therefore, advanced lifters need to focus on their gym progress and not trying to drive scale weight up too quickly, or else they'll just end up getting fat. In fact, if you are a lifter past the intermediate stage, you don't necessarily need to see the scale going up on a regular basis to ensure you are making progress (more on this to come).

So how would we implement these guidelines into our diet?

Again, remember from earlier that roughly 500 calories per day is estimated to put on about a pound (~0.5 kg) of fat per week. But, what about the caloric value of muscle? This is where it gets interesting (and confusing).

A pound (~0.5 kg) of hydrated muscle is actually only about a third protein, the rest being minerals and water. So, there are only ~800 calories in a pound (~0.5 kg) of muscle [1]. However, while fat can be very easily stored at almost no metabolic cost to the body, this is not true for muscle mass. Muscle mass is much more metabolically costly to synthesize than body fat.

Also, the reality is that even when doing things right, for lifters who are not overweight, and not novices, they're going to be putting on both fat and muscle in most cases when gaining weight. Combining this fact and the metabolic cost of creating muscle tissue, the "3500 calorie rule" actually holds up decently well for setting up your diet for weight gain as well as weight loss [14]. This is not to say that a pound (~0.5 kg) of muscle mass has 3500 calories in it, but rather, that you may have to increase your calories more than expected to gain muscle because of the energetic cost of synthesizing this muscle. Essentially, the process of building muscle increases caloric expenditure and this along with NEAT may be why in some cases weight gain is much less than expected after a caloric increase [15].

Metabolic Magic?

Interestingly enough though, because of the difference in the energy value of fat and muscle (and glycogen for that matter), in some cases strange things can happen. For example, gaining a small amount of weight in a deficit, losing small amounts of fat in a surplus, or even gaining small amounts of fat in a deficit are all technically possible because of the differences in the water and energy contents of different tissues of the body.

To illustrate how these scenarios are possible, let's take the example shown in a recent study of overweight elderly men who had experienced a significant amount of age-related muscle atrophy who began a resistance training program and a protein supplementation regimen [16].

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One of the groups in this study gained 1.3 lbs (0.6 kg) of bodyweight on average after 16 weeks. To get to this change in weight, they lost 2.4 lbs (1.1 kg) of fat mass, while gaining 3.7 lbs (1.7 kg) of muscle mass on average. So, they must have been in a surplus because of the gain in bodyweight right? Believe it or not they were actually in a deficit of about ~65 kcals per day!

How is this possible and how can I know that? Well, the amount of energy your body liberates from 1 lb (~0.5 kg) of pure body fat by burning it to "replace" an energy deficit is ~4270 kcals. Likewise, if you are in a surplus of ~4270 kcals and you generate no lean body mass and only fat, that will result in the gain of 1 lb (~0.5 kg) of pure body fat. I know you're wondering "Why isn't the value 3500 kcals?" Well in this case, that is because 3500 kcals is the value of 1 lb (~0.5 kg) of adipose tissue, which is comprised mostly of pure fat, but also contains water and some minerals. When tracking bodyweight changes at home, the "3500 calorie rule" is a good estimation tool, but lab measurements track changes in pure body fat rather than adipose tissue, and a pound of pure bodyfat therefore has a higher energy content.

Changes in lean mass however carry a much smaller energy value. When using a body composition measuring device such as a DEXA scanner, lean mass changes include their respective water contents (and in a 2 compartment model, measuring only fat mass and fat-free mass, water is considered fat-free mass). As I previously mentioned, muscle mass is in large part water, and on top of that, protein (which is what muscle is made of) has less than half the energy per gram than fat does (~4 kcals/g vs ~9 kcals/g, more on this to come next level). Thus, to gain a pound (~0.5 kg) of muscle requires ~830 kcals to be "deposited" in the body (only ~1/5th the energy content of pure body fat).

So what that means is that despite gaining 1.3 lbs (0.6 kg) of total weight, the subjects were in a net deficit of 7300 kcals over the course of the study as it requires a surplus of only ~3100 kcals to build 3.7 lbs (1.7 kg) of muscle while a loss of 2.4 lbs (1.1 kg) of fat meant a ~10400 kcals energy deficit was "replaced" by burning body fat (10400 kcals –3100 kcals = -7300 kcals). Meaning over the 16 weeks (112 days) of being observed, the participants gained weight while being in a ~65 kcals daily deficit (-7300 kcals divided by 112 days is 65 kcals/day)!

Granted, this is an extreme example. Overweight individuals can lose fat at a rapid pace because there is more of it to mobilize [17], people who are untrained make rapid gains in lean mass [18], and age-related muscle atrophy is essentially detraining; and detrained lifters make rapid progress when performing resistance training until they get back to baseline [19]. Thus, in rare situations like this it is in fact possible to gain weight in a deficit because there are less calories in the same weight of muscle tissue (due to the water content and energy density) compared to fat tissue. Thus, the total weight of hydrated muscle gained can be greater than the weight of body fat lost, and because the energy value of the total fat lost is greater than the muscle gained, weight gain can actually occur in a deficit.

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For similar reasons, a second group in the same study [16] actually lost a small amount of fat ~0.2 lbs (0.1kg) while in a very small caloric surplus, and concurrently gained ~4.2 lbs (1.9 kg) of muscle (and water).

Like this odd scenario of losing fat in a surplus, other odd scenarios like gaining fat on a deficit are also potentially possible. For example, if someone well trained ceased training all together, was bed ridden due to injury, and started eating a low-protein diet in a slight deficit, conceivably over time they could gain a small amount of fat while making up for the deficit primarily by metabolizing large amounts of lost lean mass for energy. It would not be unrealistic in this scenario for a lifter to lose say 13 lbs (~6 kg) of muscle and gain 2 lbs (~0.9 kg) of fat over a 16 week period. If this occurred, the hypothetical individual would have actually been in a small deficit on a day to day basis despite the slight gain in body fat.

So why am I telling you this? What purpose does this serve besides confusing you?

The main point is that while you should use bodyweight changes as surrogate values to help you adjust your diet, don't get so hung up on the numbers mathematically - the "3500 calorie rule" is actually a huge simplification. Don't get me wrong, it's an important and useful simplification, but you should understand its limitations.

You might also be wondering when this would ever apply to you (since you likely are not an overweight and sedentary sarcopenic 65 year old who is lifting weights for the first time)? Well, there are some instances where this "metabolic magic" might actually affect you. For example, if during a diet break you visually get leaner while gaining weight, you might be gaining more weight from regained glycogen and muscle tissue (and the gut weight of more food) than the small amount of weight that you are losing from body fat. In this case, you would likely actually be gaining weight while still being in a slight deficit. This scenario could also occur while slowly increasing calories after a cutting phase.

It's also important to point out that these "body recomp" changes, where slight deficits or surpluses produce changes in both muscle mass and fat mass simultaneously, happen slowly. Even in the study example I used with overweight elderly men training for the first time, the changes occurred over a 4 month time period. It's not as though large amounts of muscle can be gained while large amounts of fat are lost in a few short weeks, especially in well-trained individuals. This is why dedicated periods of purposeful fat loss and muscle gain are recommended.

However, advanced drug-free individuals simply can't gain large amounts of muscle mass in short periods of time. Therefore, the knowledge that the body can slowly gain muscle over long periods of time even without enforced calorie surpluses should give advanced lifters confidence in a slower lean gaining approach.

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Practical Differences Between Weight Loss And Weight Gain

So, what it all boils down to is that the real difference between weight loss and weight gain is simply that the rate of weight gain should be slower to avoid gaining excessive body fat.

Thus, to enter a muscle building phase you establish a caloric increase per the relationship between your target rate of weight gain based on your training age and its relationship with the "3500 calorie rule". Now you don't need to enforce this caloric increase every day. Calories can vary between days, but this needs to be your average caloric increase from maintenance on a day to day basis over time.

Below are the intakes for a 180 lb (82 kg) male and a 130 lb (59 kg) female respectively at each training age, calculated by multiplying 3500 by the target rate of gain per month, divided by 30 to show the daily average increase required:

Experience Level:	Calories Above Maintenance at 180 lbs (82 kg)	Calories Above Maintenance at 130 lbs (59 kg)
Beginner	1-1.5%/month = ~200-300 kcals/day	1-1.5%/month = ~150-225 kcals/day
Intermediate	0.5-1%/month = ~100-200 kcals/day	0.5-1%/month = ~75-150 kcals/day
Advanced	Very slight increase up to 100 kcals/day	Very slight increase up to 75 kcals/day

And lastly, just to be clear on how we would be tracking these changes, let's use an intermediate 180 lb (82 kg) male as an example. If you have a goal of gaining 1 to 2 lbs (0.4 to 0.9 kg) per month, this simply means that we would divide that by 4 to get a weekly target.

So a beginner at 180 lbs (82 kg) trying to gain up to ~3 lbs (~1.3 kg) per month, might aim to gain ~3/4 of pound (~0.3 kg) per week, the intermediate might be about half a pound (~0.2 kg) per week, and advanced will basically be maintenance or slightly higher. And that's essentially how you run a lean bulk based on your body mass and experience level.

As previously discussed in the Mindset & Materials portion of this guide, the weekly averages of daily weigh-ins will be your most helpful tool to know whether your nutrition is helping you progress towards your goals or not. If you are not gaining weight or losing weight at the targeted pace, you can adjust your intake based on the "3500 calorie rule" to reach your target. Having this first level under control is honestly about 70 to 80% of the battle when it comes to getting your diet in check. The subsequent pieces of The Pyramid combined will make up the rest.

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Level 2

MACRONUTRIENTS & FIBER

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With our caloric goal figured out, the next step is to figure out where those calories come from. Our three main calorie providers are protein, carbohydrates, and fat. We will continue to refer to these as your macronutrients, or "macros".

Yes, to be technical, alcohol is a fourth calorie provider. However, we're going to assume that this is not a regular staple in your diet, especially when your lifestyle is heavily aimed towards fitness goals. Alcohol measurement and tracking will be considered toward the end of this guide, but it definitely does not belong here in Level 2 as part of such a vital topic.

In this section, we will go over the function of the macronutrients and then establish macronutrient intakes that are appropriate for most athletes in various conditions. Also, we will introduce some alternative guidelines for those who may fall outside of the norm, and then discuss how to identify where you fall within the spectrum of all of these values. To close it out, we will go over fiber intake recommendations for health and nutrient absorption.

How To Prescribe Macronutrient Guidelines

Macronutrients are often prescribed either based on bodyweight or based on a percentage of calories. Both of these methods have their pros and cons.

Macronutrient prescriptions based on grams per pound (or grams per kilogram) are relative to your body mass, which for the most part makes sense. Typically the larger you are, the greater your energy requirements will be. However, this method does not give respect to how much of a "caloric budget" you might have. If you happen to have a relatively low or high energy expenditure for your body mass and you determine all three macros based purely on your bodyweight, you could very well eat so much or so little that you'd be inadvertently gaining weight or losing weight faster or slower than is appropriate, subverting the goals of Level 1.

This is the main benefit of using a percentage of calories-based prescription to dictate macronutrient intake; a percentage is always scaled to how many calories you are consuming. If you followed the guidelines in Level 1, your caloric intake will be based on the appropriate rate of weight gain, weight loss, or maintenance that you calculated.

So, there is value in using both methods. Macronutrients scaled to bodyweight take into account your mass, which is especially important to consider in the case of protein. Protein is not primarily used as a source of energy (compared to fat and carbohydrate), but rather for the build and repair of tissues. For this reason, depending on the macronutrient in question, I prefer to use a mix of these two methods.

As I stated, protein's primary function is to build and repair the structure of our tissues. At most (depending on what we are doing), 10% of our energy might come directly from amino acids (the building blocks of protein). Additionally, the liver can convert protein that is not used for structural purposes to glucose, ketones, and other metabolic

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substrates which are the body's main sources of energy. That said, even combining the amount of protein that is used directly for energy with the amount that the liver converts to glucose, ketones or other substrates, the role of protein as an energy source pales in comparison to that of carbohydrate and fat [1].

Due to these functions, I calculate protein relative to bodyweight, with fats and carbohydrates based on a percentage of calories. This allows protein to be scaled to your body mass as its role is primarily structural, while carbohydrate and fat are scaled to your total energy allotment since their primary role is to provide energy.

The other information we need to know is that there are ~9 calories per gram of fat, and then ~4 calories per gram for both carbohydrate and protein. Those values, combined with our daily calories from Level 1 of The Pyramid, will help us assign our macronutrient intakes.

	Protein	Carbohydrate	Fat
Calculation Method	Gram per pound of bodyweight	Percentage of total calories	Percentage of total calories
Calories per Gram	~4 calories	~4 calories	~9 calories

As with each level of this guide, we will go over these calculations for those who are dieting, and those who are gaining.

Setting Macros For A Fat Loss Phase

Although the recommended macronutrient ranges that are acceptable for dieting and gaining are going to be similar, there are some subtle and important differences. When in a cutting phase of any kind, you're going to have to "rob Peter to pay Paul", so to speak - you'll be forcing yourself to either not get enough carbs, protein, or fat simply because you don't have enough calories in your budget. Therefore, we need to be a bit more cognizant of how our macros are set up during a dieting phase to protect against potential lean body mass losses, and also to improve adherence and maintain gym performance.

However, if you've already screwed up Level 1, you will inevitably have already screwed up your macronutrient distribution as well. For example, if you have ignored the advice from Level 1 and decided to try to lose 1.5-2.0% of your bodyweight per week, you have to eat fewer calories and thus you have less calories to distribute among your macros. You may be thinking "that's alright I can just eat a high protein diet to offset this", but you would be forgetting that protein doesn't fuel your training, and lifting weights is the primary weapon you have against lean body mass losses. Therefore, someone who is eating a lower protein intake than I would recommend, but who is dieting and losing 0.5

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to 1% of their body weight per week, is likely going to maintain more muscle mass than someone eating as much protein as I recommend while losing 1.5 to 2% of their body weight per week. So, remember that the rank of importance of each level continues to matter. Additionally, protein can only do so much to protect against losses of lean body mass while dieting.

Setting Protein Intake

As I mentioned earlier, protein contributes a small amount to energy expenditure. During a diet this amount increases for several reasons. First, greater levels of activity increase protein requirements [2] and activity goes up when cardio is performed in addition to weight training during a diet. Additionally, protein is used more for fuel because less fat and carbs are available due to caloric restriction [3]. Finally, with less dietary carbohydrate and fat available, more glycogen and body fat are utilized; as their stores decrease, this forces the body to rely more on protein as an energy source [4].

Both dietary and body proteins can be used for energy in the face of this energy shortfall. Thus, the body increases its anabolic response to protein to protect against lean mass losses, becoming more efficient in its use of protein [5]. However, this increase in efficiency often does not prevent losses of muscle mass, especially in lean individuals even when weight training [6]. For this reason, it has been hypothesized that consuming a higher intake of protein in the range of 0.8 to 1.2 g/lb (1.8-2.7 g/kg) while dieting could offset losses of body protein to help preserve muscle mass [6-8].

However, this hypothesis has not been comprehensively tested and proven. At this stage, there are studies clearly showing the superiority of higher protein intakes for lean mass preservation in non-overweight resistance-trained populations when comparing:

- 0.72 g/lb (1.6 g/kg) to 0.36 g/lb (0.8 g/kg) for 1 week [9]
- 1.05 g/lb (2.3 g/kg) to 0.45 g/lb (1 g/kg) for 2 weeks [10]
- and 1.09 g/lb (2.4 g/kg) to 0.55 g/lb (1.2 g/kg) for 3 weeks [11].

However, these comparisons of moderate versus low, or high versus low protein intakes don't actually confirm the hypothesis that high protein intakes are more beneficial than moderate intakes while dieting. To date, only three studies have directly compared moderate intakes to high intakes during an enforced caloric deficit.

- In 2013 my colleagues and I found that while body composition after dieting was not found to be different while consuming 1.3 g/lb (2.8 g/kg) compared to 0.72 g/lb (1.6 g/kg) after 2 weeks, the higher protein group did report lower levels of fatigue, athlete related life stress and diet dissatisfaction [12].
- Additionally, researchers performing a study that was published around the same time found that while body composition after dieting was not found to be different while consuming 1.09 g/lb (2.4 g/kg) compared to 0.72 g/lb (1.6 g/kg) for 3 weeks, acute markers of protein synthesis were more elevated in the higher protein group [13].

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- Most recently, researchers reported that after 8 weeks of energy restriction a pre and post workout whey-supplemented group maintained more muscle mass and gained more strength than a pre and post workout carbohydrate-supplemented group, which gained more muscular endurance. On average the whey-supplemented group consumed 1.11 g/lb (2.46 g/kg) of protein per week, and the carb-supplemented group consumed 0.98 g/lb (2.16 g/kg) [14].

Unfortunately, none of these studies are perfectly equipped to answer the question at hand.

In our study we used a well-trained anthropometrist to track changes in skin folds, and while this has adequate precision for tracking actual skinfold thickness changes (measuring body fat), estimations of lean body mass changes have a high degree of error. Also, the second study listed was not on resistance trained participants, and the training they performed was not progressive resistance training, so the results can only be extrapolated so far.

Finally, the last study assessed well-trained lifters performing realistic training for 8 weeks, and used a reliable method of assessing body composition (hydrostatic weighing), but it did have one confounding variable. In this study, not only was a high protein intake compared to a moderately high intake, but also nutrient timing differed between groups. The whey supplemented group consumed the additional protein around training, which could have influenced the outcomes.

So while it is difficult to say with absolute certainty that a higher protein intake will preserve more lean body mass during a diet, what can be said with certainty is that a protein intake in the range of 1.1-1.3 g/lb (2.5-3 g/kg) would not be harmful [15], and that there is evidence that protein intakes in this range (actually higher) appear to be more satiating as they tend to reduce energy intake [16, 17] and may positively affect mood state compared to lower intakes [12].

To conclude, based on the above evidence and my experience as a coach, what probably makes the most sense for protein intake while you're dieting is somewhere between 1.1 to 1.3 grams per pound (2.3-2.8 g/kg) of body weight.

Setting Carbohydrate & Fat Intake

Once protein is set, the next step is to calculate carbohydrate and fat intake.

Now carbohydrates are not considered to be "essential" but they will certainly benefit your training, and lifting weights is the number one thing you can do to prevent lean body mass loss. Number two would be your rate of weight loss, and number three would be your macronutrient intake that we're figuring out right now. If there is one thing that this Nutrition Pyramid tends to under-represent is how important your training is to your muscle and strength goals. (If you haven't purchased it already, I would highly advise getting my companion book on training.)

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That said, while dietary carbs are important for performance, they aren't technically "essential" for life and function. Therefore, I always like to set fat first.

While you're dieting, I'll lower the fat a little more than I would typically in the off-season, just because it allows for a higher carbohydrate intake and if carbs are too low during energy restriction, it can sabotage your training [9, 18-20]. This would then in turn degrade your ability to retain lean body mass.

So with fat intake, I recommend 15-25% of calories while you are dieting for most people, and then the remaining calories are assigned as carbs. Yes, 15% is lower than ideal in most cases, but that's why diets don't last forever. Additionally, we'll discuss a physiological minimum based on bodyweight in a moment. But, keeping fat a touch lower while dieting allows you to consume more carbohydrate and it's important to retain enough carbohydrate so you can sustain your training, and in turn so you can sustain your muscle mass. I will go over an example of these calculations after we discuss the appropriate values for non-dieting individuals.

Minimum Recommended Carbohydrate & Fat Intake

Finally, it should be noted that some minimum intakes should be established for carbs and fat since they are your primary energy sources, and in the case of fat, it is an essential nutrient. Dieting requires caloric restriction, and caloric restriction requires macronutrient restriction, but for those with a very low energy expenditure it is possible that by using the guidelines in this book you may end up inappropriately low in your intake of fat or carbohydrate. To prevent this, use the minimum intakes of 0.25 g/lb (~0.5 g/kg) for dietary fat, and 0.5 g/lb (~1 g/kg) for dietary carbohydrate.

Setting Macros For A Gaining Phase

Athletes who are attempting to add muscle and strength over time don't have the same macronutrient needs as dieters. Since they are in an energy surplus or at the very least at maintenance, and likely have higher body fat and glycogen levels than a dieter, the rationale we previously discussed for having as high of a protein intake no longer applies. For the most part, research has found that the beneficial anabolic effects of high protein diets don't extend past ~0.8 g/lb (~1.8 g/kg) [8]. In fact, only in a few instances is there a trend in the research for higher protein intakes to show benefits for muscle gain [21-23] and resistance training performance [23-26]. However, these statistical trends are inconsistent and sometimes confounded by differences in nutrient timing. For that reason, a protein intake of 0.8 to 1.0 g/lb (1.8-2.2 g/kg) of bodyweight will likely maximize any potential benefit of a high protein diet (you could go higher for satiety purposes, but that can actually be a barrier for some people who get too full in the offseason).

Now that value may prompt some of you to freak out. I know I just told you something below a gram per pound, which in the bodybuilding community is often seen as the absolute minimum intake. Before you dismiss me as an egghead with no in-the-trenches

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experience, keep in mind that I'm a bodybuilder who has achieved reasonable success in natural bodybuilding, I coach bodybuilders who have competed at the highest level in natural bodybuilding, I've conducted research on bodybuilders and dietary protein, and I've provided you with plenty of references to support these numbers. So please try not to throw your protein shake at the screen as you read this.

Think of it this way, you've got the calories to support you when gaining. Part of the reason why dieting theoretically requires more protein, is because more protein is used for energy when you're leaner and restricting calories [4]. These conditions are not present during a surplus, so therefore you simply don't need as much protein and you'll benefit more from eating higher amounts of carbohydrate and fat.

Speaking of carbohydrate and fat, since a gaining phase gives us more calories to play with, and we don't have to worry about our carbohydrate intake being too low to sustain our training, we can have a higher percentage of our calories coming from fat. For most people I recommend 20-30% of total calories, and once again, the rest is assigned as carbohydrate.

The table below will help to sum up our cutting and gaining scenarios for subsequent calculations.

MACRONUTRIENT RECOMMENDATIONS	PROTEIN	FAT	CARBOHYDRATE
CUTTING	1.1 – 1.3 grams per pound (2.3 – 2.8 g/kg) of bodyweight	15 – 25% of total calories per day	The remaining calories to fulfill daily requirements
GAINING	0.8 – 1.0 grams per pound (1.8 – 2.3 g/kg) of bodyweight	20 – 30% of total calories per day	The remaining calories to fulfill daily requirements

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Example Gaining Calculations

Let's stick with our 200 lb (~90 kg) male from the last example and take a fairly average energy balance for someone of this size. At this bodyweight (recalling Level 1), this individual will probably need somewhere between 2600 and 4400 calories to maintain their current weight.

To keep things nice and even for our calculations, let's go with 3000 calories per day as an intake that would elicit no changes in bodyweight. If we would like to take this person into a lean gaining phase as an intermediate lifter, we could add 200 calories per day to create a slight surplus (refer to the recommendations made at the end of the calorie setting chapter). Our goal intake for this individual is now 3200 calories per day.

Let's start with 1 gram of protein per pound (2.2 g/kg) of bodyweight, which equals 200 grams.

200 grams of protein x 4 calories per gram = 800 calories

We will then set fat at 25% of his total calories for the day, which ends up as roughly 90 grams.

3200 calories x 0.25 = 800 calories from fat

800 calories from fat ÷ 9 calories per gram = 88.89 grams of fat

And lastly, carbohydrate intake will simply be based on the remaining calories that have not already been used by protein or fat, which comes out to approximately 400 grams.

800 calories from protein + 800 calories from fat = 1600 calories left for carbohydrates

1600 calories of carbohydrate ÷ 4 calories per gram = 400 grams of carbohydrate

To sum it up, this 200 lb (~90 kg) athlete in a lean gaining phase could set up a daily macronutrient intake goal of 200 grams of protein, 90 grams of fat, and 400 grams of carbohydrate.

Example Cutting Calculations

Now if this same individual is dieting, these numbers will be slightly different because he needs to drop his daily calories down enough to lose 0.5 to 1.0% of bodyweight per week.

With the same maintenance value of 3000 calories, and using the "3500 calorie rule" of losing approximately a pound per week (~0.5 kg), let's bring him 500 calories down from his day to day intake and then add some cardio expenditure on top of that (which we won't show here).

He's now consuming 2500 calories, and we'll bump that protein up a little bit. Let's say around 1.1-1.2 grams per pound (2.5 g/kg), to 225 grams.

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225 grams of protein x 4 calories per gram = 900 calories

We'll put fat around 20%, which is roughly 55 grams per day which is still over the 0.25 g/lb (0.5 g/kg) minimum of 45-50 grams.

2500 calories x 0.2 = 500 calories from fat

500 calories from fat ÷ 9 calories per gram = 55.56 grams of fat

And now we have 1100 calories left over for carbs, which ends up around 275 grams.

900 calories from protein + 500 calories from fat = 1100 calories left for carbohydrates

1100 calories of carbohydrate ÷ 4 calories per gram = 275 grams of carbohydrate

To sum it up, this 200 lb (~90 kg) athlete at the beginning of a fat loss phase could set up a daily macronutrient intake goal of 225 grams of protein, 55 grams of fat, and 275 grams of carbohydrate.

This is enough carbs to satisfy training, enough fat to not lose his mind and maintain some resemblance of a normal diet, and enough protein to protect against lean body mass losses - a very sound, well-constructed starting point.

Other things might come into play like refeeds or diet breaks, but this will serve as the basic caloric and macronutrient set up. We will talk about those topics, along with how to "hit" these numbers, later on in this guide.

Who Should Use These Recommendations?

These numbers are not going to be perfect for everybody. In fact, for some they could get in the way of progress.

I probably have 10-15% of my clients on percentages of fat that are higher than 30%. I also have another maybe 5% who are actually on borderline ketogenic diets, meaning that they're consuming a daily average carbohydrate intake somewhere around 80-120 grams or lower per day, which might even be while they're not dieting. But like I said, this is not the majority; at most it represents 20% of the folks I work with. This means that maybe 2 out of 10 people shouldn't have their carbs as high as the earlier calculations would estimate. These people happen to do better with a high-fat, low-carb approach instead.

So, how do you figure out if that's you?

Although there are a few genetic indicators that can help us guess what type of diet would best-suit you, you're probably going to have to test this out based on some self-trial methods that I will discuss in the next section. But the above calculations will most likely result in effective macronutrient distributions for the majority of folks who are exercising, not overweight, and who are still young through middle aged.

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Now, it is well known that exercise and maintaining a healthy bodyweight improves your ability to metabolize carbohydrate, but why is age a factor?

Well, most older people who struggle with higher carbohydrate diets do so because they have lost muscle mass and gained fat mass as time has passed, resulting in insulin resistance. However, even for the folks who have picked up fitness later in life or who are still committed to their resistance training, some independent effects of aging do negatively impact the body's ability to handle carbohydrate [27].

Also, protein doesn't continue to give the same anabolic response that it used to as you age [28]. So although your fat and protein might need to increase over the years, just remember that this occurs alongside an overall reduction in energy expenditure [29]. This means you won't have as many calories to play with overall, which is why you don't want to go so low on carbohydrates that you're actually getting ketogenic unless you find that actually works well for you.

So again, I will say that the recommendations I made are a great place to start for 80% of people who are not overweight, lifting weights regularly, and who have not yet started the final third of their life span (arbitrary number here, but let's say 60 years old). So, if you are in the first ~2/3rds of your lifespan, not overweight, and regularly lifting weights, the recommended ranges will more than likely "work" well for you. For everyone else, I wish I could make a guide that would give you the perfect macronutrient ratio along with a unicorn to ride across the magic dietary rainbow, but the reality is that nutrition is just more individual than that, and you'll have to bear with me on that limitation.

But How Do I Know If A Higher-Fat, Lower-Carb Diet Is Right For Me?

Although it is not very typical among recreational and competitive bodybuilders and strength athletes to fall into this category, I think it is important to address how to tell if you are someone who would do better with a higher fat to carbohydrate ratio.

Now when I say "high fat", I typically am referring to a fat percentage higher than 35% of total calories [30] as opposed to the aforementioned 15-30% recommendation (depending on whether you are dieting or gaining). In turn, keeping calories and protein the same, this fat intake would lead to a generally lower carbohydrate diet compared to my earlier recommendations. And when I say "lower carbohydrate diet", I'm referring to an intake that may approach as low as 0.5-1.5 g/lb (~1-3 g/kg) of bodyweight.

I am not necessarily referring to a ketogenic diet, which is extremely low in carbs, sometimes defined as 50g or lower [31]. These diets necessitate extremely high fat intakes to maintain calorie balance, often up to or above 60% of total calories. These "keto" diets have become quite trendy and popular as of late along with the idea that eating more dietary fat helps you to burn more fat as fuel, but that's not exactly how it works. Sure, you shift your body's fuel usage more towards burning fat, but since you are also consuming more fat, it doesn't necessarily result in greater fat loss [32].

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Additionally, going too low in carbohydrate can potentially degrade performance in the gym. In reality, a high-fat diet is not potentially useful because high fat is somehow beneficial, but rather because some people have difficulty utilizing carbohydrates. In conclusion, there is a time and a place for a high-fat, low-carb diet, and it is most often found in people who are insulin resistant [33].

So, how do you know if you fall into this camp and could benefit from that type of diet?

Well, there are a few predictors that may indicate you have a greater likelihood of being insulin resistant, and there is a way to test this hypothesis yourself using another set of basic calculations and subjective data tracking that I will outline in this section.

The likelihood that you will do better with more dietary fat due to insulin resistance is typically increased with age [27], a family history of diabetes [34, 35], the presence of a condition in women called polycystic ovarian syndrome (PCOS) [36], or oligomenorrhea [37], which is represented by a menstrual cycle that occurs less frequently, specifically taking longer than 35 days to recur. While these first two factors can sometimes be affected by bodyweight and exercise, women with PCOS or oligomenorrhea typically have higher androgen levels (male sex hormones) than most women which predisposes them to be insulin resistant to some degree [37] regardless of activity or body-fat levels.

Additionally, it is not at all uncommon for women with higher androgen counts to be disproportionately represented in athletic populations. This is even more so true in strength and power sports, which tend to have more women with higher androgen levels even compared to other sports [38]. But, don't despair if you need to modify your diet due to potential insulin resistance if you are a woman with PCOS or oligomenorrhea. The upside is that you likely have a higher than normal androgen count, which may mean you potentially perform better than other athletes [39]. To further allay any fears, take heart that there is pretty convincing evidence that a higher-fat, higher-protein, lower-carbohydrate diet can be very effective for producing fat loss for those with insulin resistance [33, 40-42].

Now, what if you are a male who does not respond well to carbohydrates or a female who does not have a diagnosis of PCOS or oligomenorrhea but still doesn't do well on a high-carbohydrate diet? How do we test if a higher-fat diet would be better?

In an ideal world, you could simply go get some blood tests done to determine your insulin sensitivity, but that is not a typical option for many people. And even then, if a doctor finds you to be insulin resistant, the typical recommendations are to lose some weight, do some resistance training, and get active. Well, if you are reading this text you are probably already taking care of all of those things. So if you get that diagnosis, you're left with no solution according to the doctor.

So beyond going to the lab, I think an even better way to test your response to a higher-fat, lower-carbohydrate diet would be to run some off-season testing and data collection on yourself. I would recommend taking about a month to consume a 40% fat

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diet, with the same protein and calories that you have normally been consuming. This would simply be changing the carb to fat ratio and nothing more.

Throughout this month, write down and record a 1 to 10 rating on mood, energy, and training quality each day. Mood and energy are pretty straight forward to define, but I'd like to clarify that gym quality is different than gym performance. We are not looking at your program, volume, or poundage of weights lifted (you can track this as well, but only if you keep your training approach the same). This is more of how you perceived your overall effort and mental state during the day's training. After about 4 weeks of tracking mood, energy, and gym quality, take an average score of those ratings.

When that month is complete, take another month consuming a 20% fat diet while keeping calories and protein the same. Once again you will track the same data: food and bodyweight (as always), and also your 1 to 10 ratings for mood, energy and workout quality (and again, also a good idea to track gym performance, but you would need to keep the overall training structure the same). I would recommend running these trials at least twice each, making this a 4-month commitment. At the conclusion, you'll be able to see if your ratings were higher for one diet or the other, and you'll have your answer.

This might seem like a big pain in the ass to find out whether you do better on a higher-fat or higher-carb diet, but in the off season it's definitely worth it. If you see a big difference in these ratings from your 40% fat months and your 20% fat months, you might be able to conclude that a higher-fat diet is something for you to stick to in the long term. Also, put it in perspective, if you plan on lifting weights and trying to improve your body composition for the rest of your life while you're able (which if you're reading this book I hope you are), then 4 months is just a drop in the bucket!

Fiber

The last topic we'll cover here in Level 2 is our fiber intake, which is important for gut health and nutrient absorption. It can also effectively decrease your energy intake because not all fiber has a calorie value, yet it "counts" as a carbohydrate and may increase satiety [43].

I would say that a healthy minimum fiber intake would be 20 grams for females and 25 grams for males per day. As a maximum for both genders, I'd say we don't need to go above 20% of your total carbohydrate intake on any given day. Now it might surprise some people, especially some old school bodybuilder types, that we even have a maximum fiber intake at all. But if you only eat oats and clean foods to increase this number well above recommended values, it can actually be detrimental to your GI health and your ability to absorb some nutrients [44], while also potentially giving you a very bloated look if taken to the extreme.

Another way to go about estimating healthy fiber intake is to give it a blanket minimum rule of 10 grams per 1000 calories. This is probably more practical for individuals who are on a ketogenic, low-carbohydrate, or low-calorie diet. Therefore, for example if you

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are consuming 1500 calories you would set a minimum intake of 15g of fiber, or 20g if you were consuming 2000.

There is also the question, “should I count fiber towards my carbohydrate total?”

Traditional nutrition often teaches that fiber is not digestible by humans, but in fact some fiber, while not digestible in the small intestine, is fermented in the colon and that by product provides energy. However, determining which fibers provide calories and exactly how many is quite difficult [45]. Because of this, my basic recommendation is to simply count your fiber as carbohydrate, and to make sure you fall between the minimum and maximum ranges I’ve provided here.

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Level 3

MICRONUTRIENTS & WATER

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Just to set the stage a little bit, macro means big and micro means small. Thus, you need macronutrients in much greater amounts compared to micronutrients.

So far we have only discussed the macros which have been presented in the amounts of full grams. In contrast, when we talk about micronutrients, we are talking about quantities that are very small, from milligrams on down. The occasional time a micronutrient is required in gram level amounts, it is needed in amounts of maybe a couple of grams at most.

That said, these nutrients that we only need in tiny amounts can make a big difference in general health, gym performance, hunger levels, mental state, and overall ability to put on muscle if we happen to have deficiencies in them. This section will go over the types of micronutrients and how to use your food choices to fulfill their daily requirements. We will then finish Level 3 with fluid intake recommendations and how to tell if you are consuming enough to keep yourself hydrated and functioning properly.

Identifying Our "Micros"

There are two broad categories of micronutrients in nutrition. We have our minerals, which are inorganic, and then we have our vitamins, which are organic. I'm not going to go through a full-blown nutrition lesson here on every single micronutrient and all of their individual purposes, but feel free to do a simple Google search to find all of that information on your own time if interested.

However, for completeness I am going to go over what they are and the different categories because I think that will give you some perspective on why you can't (or why you *shouldn't* in my opinion) just stop at your macronutrient intake without any consideration of other nutrients. Once we cover the categories, we'll talk about how to apply that awareness into your diet and make sure all of your nutritional bases are covered.

Minerals

So to start, minerals are those inorganic things that are on the periodic table. We have macrominerals, meaning those that our body needs in greater abundance, and trace minerals which we need in smaller amounts.

Your macrominerals include calcium, phosphorus, potassium, sulfur, sodium, chlorine and magnesium. Those of you with a nutrition or chemistry background might notice that a lot of those are the electrolytes, which tie into our fluid intake and balance.

The trace minerals, which serve important functions in human physiology even though they are not required in large abundance, are elements such as iron, copper, cobalt, zinc, molybdenum, iodine, and selenium. There are others, but those are some of the more prominent ones.

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Vitamins

Vitamins, our organic micronutrients, also come in two different types: fat soluble and water soluble. These category names refer to the way that these different vitamins can be absorbed into the body.

Fat soluble vitamins are absorbed with the assistance of fat via the small intestines and tend to stick around in our bodies much longer than their water soluble counterparts. Water soluble vitamins are quickly excreted through our sweat and urine, which is why we typically need to consume them more often.

The fat soluble vitamins are A, D, E and K, easily remembered with the acronym "ADEK". Once you have memorized ADEK, then you just need to remember that all the rest of the vitamins are water soluble. The water soluble vitamins consist of the 8 different forms of vitamin B and also vitamin C. Once again, memorizing the chemical name for each, its function, and what occurs if you are deficient or consuming excessive doses is not important in the context of your goals and therefore not appropriate for inclusion in this guide. However, what I *do* think is important for our discussion are the differences between fat soluble and water soluble vitamins and how their absorption nature will affect how often we need to ingest them.

Because we drink and urinate all day long, we have a large amount of water processing through our body constantly. Thus, water soluble vitamins need to be replenished on a regular day-to-day basis. So it is harder to overdose on a water soluble vitamin because you can urinate and get rid of the excess. However, it is easier to be deficient in a water soluble vitamin since we have such a high turnover rate.

The opposite is true for fat soluble vitamins. It is easier to overdose because they stay in our body longer, and thus they are also harder to get rid of. And likewise, it is more unlikely that you will be deficient unless it's just a consistency issue. It takes longer for a deficiency to manifest in the fat soluble vitamins.

So that is the big picture background information that is good to know moving forward. Next we need to think about how we ensure that our diet is set up so that we are consistently consuming an appropriate intake of micronutrients.

Implementing The "Inclusive" Vs. " Exclusive" Mindset

You'll recall from Level 1 that your dietary mindset and outlook are the key ingredients to creating a nutrition plan that will help you progress toward any of your size or strength goals. We have to make sure we *include* foods high in micronutrients rather than trying to *exclude* foods that aren't high in them.

This "inclusive vs exclusive" mindset is especially important while dieting.

When you are eating less food, you will also be eating fewer micronutrients. Many popular diets, even ones that sound like they have adequate variety and inclusion of

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various micronutrient-dense foods, are actually micronutrient deficient [1].

The traditional approach to dieting of focusing on "clean" foods and avoiding "dirty" foods popularized by bodybuilders, which supposedly increases the micronutrient density of a diet, can often in fact be so restrictive as to leave out key nutrients and result in micronutrient deficiencies [2-6].

Deficiency Considerations When Dieting

The five deficiencies most consistently reported by dieting bodybuilders from surveys in the 1980's and 1990's were vitamin D, calcium, zinc, magnesium, and iron [7]. Sometimes, these deficiencies can have significant impacts related to the body composition goals of the individual. We'll cover some examples here.

Zinc deficiencies can in fact result in a down regulation of thyroid to the point where energy expenditure is substantially reduced. In one case study, one subject increased their resting metabolic rate (RMR) by 194 kcals after 4 months of supplementing with zinc, and the other subject increased RMR by a whopping 527 kcals after supplementing with zinc for 2 months [8]. This is a real life example of the potential consequences of being "exclusive" in your approach to food source selection versus "inclusive" like we are recommending.

COMMONLY REPORTED MICRONUTRIENT DEFICIENCIES BY BODYBUILDERS CIRCA 1989-1994
Vitamin D
Calcium
Zinc
Magnesium
Iron

Another example of when a micronutrient deficiency can negatively affect your progress is iron, which is an especially common deficiency in female athletes (even more so when energy restricted). In one study on a group of female volleyball players who were supplemented with iron, it was found that during strength testing the iron-supplemented group displayed greater total strength across a range of barbell lifts compared to the control group that did not receive iron supplementation [9].

Clearly, in some cases correcting a deficiency during dieting can be incredibly important for success. Calcium is another common deficiency and also is required in relatively large amounts. Consuming dairy on a diet can fix this issue; and low fat, low carb, high protein Greek yogurts can be fit into diets that are low in calories, fat or carbs, and are a great way to prevent deficiency. However, for those who don't have access to this type of Greek yogurt in their location, or have a specific reason that precludes dairy

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consumption, calcium supplementation may be something to consider.

- *Zinc deficiencies can negatively impact your metabolism.*
- *Iron deficiencies can negatively impact strength.*
- *Calcium deficiencies can negatively impact bone health.*

Deficiencies can have a significant impact on health, performance, and potentially diet-success if unchecked. A diet with good variety can cover these bases, but as calories and macronutrients get lower, it will be harder and harder to prevent micronutrient deficiencies. I would advise maintaining dairy and red meat consumption (lean with fat trimmed can fit into almost any diet) and getting regular outside sun exposure (not through windows [10]) while dieting. Doing so should help you avoid calcium, zinc, magnesium, iron and vitamin D deficiencies. For those who don't eat meat, or red meat specifically, or dairy, later on in this book we'll cover supplementation which may be something to consider in your case, especially when dieting.

Deficiency Considerations When In A Gaining Phase

When you are not dieting, you'll find it much easier to prevent these deficiencies. However, you may find that you aren't as hungry, and fruits and vegetables that are high in fiber, high in water and most importantly high in micronutrients, are easy to forget about and are overlooked because of their lower calorie content. So, I would recommend consciously ensuring that you consume adequate fruits and vegetables.

The reason I am focusing on these particular food sources is because typically, the average lifter when gaining is all set in terms of having adequate amounts of the micronutrients that come from meats, dairy, and starchy carbs. But the typical place where a muscle-conscious individual's diet lacks, especially when hunger is low, is in the fruit and vegetable department.

Simple Fruit And Vegetable Consumption Recommendations

The best way to keep track of this is to ensure that you are consuming 1 serving of fruits *and* vegetables per 1000 calories that you ingest per day. For example, if you are a larger male in the off season consuming 3000 calories per day, you should aim to eat 3 servings of fruit and 3 servings of fibrous vegetables per day.

If you are a small female or lightweight male who is eating 1200 to 2000 calories per day, the aim should typically be to consume 2 servings of fruits and veggies each. However, this might be difficult to fit into your diet without going over your carbohydrate target. In this case, look around the supermarket to see what has the lowest calories per serving size. You'll find that something like berries would be a better choice than a banana for your situation.

On the flip side of that, a very large or extremely active individual who regularly consumes 4000 or 5000 calories per day might have a tough time getting that many

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calories down if he or she has to eat 5 servings of fruits and veggies daily. In this case, getting 4, 3 or even just 2 servings would suffice in order to prevent excessive bloat or fullness that could get in the way of eating adequate offseason calories. A good bare minimum intake is 2 servings of fruits and vegetables each day to prevent excessive fullness while still ensuring adequate intake; remember the order of priority, you don't want Level 3 of The Pyramid to interfere with Level 1!

Aside from extremely low or high-calorie diets, I still think that 1 serving of fruits and 1 serving of vegetables per every 1000 calories is a very good basic guideline to follow. One of the benefits from implementing this set up is that when you're dieting, these foods tend to be higher in fiber and higher in satiety [11] which will help you with diet adherence and consistency.

Produce in general tends to be very low-calorie, so you'll find that increasing your intake of these particular foods can help you feel like you're eating more despite being in a caloric deficit.

FOOD CLASS	RECOMMENDATION	NOTES
Fruits	1 serving per 1000 kcals	Round up when dieting, may not be possible if on a very low carbohydrate diet
Fibrous Vegetables	1 serving per 1000 kcals	Round up when dieting, may reduce servings when gaining if hampering caloric intake

This is exactly why when your food sources are poorly set up (i.e. eating a lot of foods that are not very rich in micronutrients), you can technically "hit your macros" with a lot of very energy-dense foods but still feel hungry and empty.

So in summary, make sure you have good food variety (especially when dieting), and make sure that you consume enough fruits and vegetables. If you do this, in addition to having well set-up macros, you will probably make progress towards your goals while getting sufficient micronutrients. Additionally, when we get to the supplementation section of this guide, we'll discuss some other ways to ensure you don't risk a deficiency while dieting.

Fluid Intake

Just like in Level 1 of The Pyramid for determining maintenance calories, there are a couple of different options for setting an appropriate amount of fluid intake for your

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body. The first method will involve calculations based off of your bodyweight, and the second method is more qualitative in nature based on your individual needs.

Bur first off, what liquids should contribute to our measurement of intake?

In this case I'd like you to include all of your fluids except for alcohol. Yes, even things we see as "dehydrating" like coffee, actually do more to contribute to our fluid intake that detract from it [12]. So all fluids, including coffee, diet soda, milk, juice, tea, flavored waters, and any drink (besides alcohol) will count towards this intake. We don't count alcohol because it is in fact dehydrating [13], and makes you urinate more fluid out than you take in.

So how much fluid should you consume?

A good place to start for those who use the imperial rather than metric system would be to shoot for 2/3 of your bodyweight in pounds, in fluid ounces. So that means if you're say 210 lbs, this equates to roughly 140 fluid ounces of liquid per day. For those using the metric system, an equivalent rule of thumb is to consume one liter of fluids for every 23 kg of your bodyweight.

Another method to ensure you are adequately hydrated is one found on the website bodyrecomposition.com. The article recommends that you should have 5 clear urinations per day, 2 of them coming shortly after your workout (or during your workout if it is long) if you trained [14]. I think this is also a perfectly fine approach for determining if your intake is sufficient, and may even have advantages over a controlled, measured amount based on bodyweight.

The reason that I tend to favor this method is that at the same bodyweight, two people can have a very different hydration status. For example, consider one person who habitually drinks a lot of fluids, and thus urinates more and has a much higher sweat rate. This person would have a very high water turnover, while someone else may not. Additionally, other things like quantity and type of training, the type and amount of cardio you're doing, and the climate you live in are all going to impact your hydration status.

As an example, let's discuss two extremely different individuals. One is a 200 lb (~90 kg) inactive sweater (doesn't sweat a lot) living in Alaska who does low-volume weight training 3 times a week in an air conditioned gym. The other is also a 200 lb (~90 kg) guy who lives is in the Middle East, is a perfuse sweater, weight trains 6 times a week, does cardio 3 times a week, and has only sporadic access to air conditioning. It's quite logical to guess that they are going to need massively different fluid intakes to maintain hydration despite having the same bodyweight.

Ensuring you are adequately hydrated is vital because even a 2% decrease in bodyweight from fluid losses can cause decreases in aerobic performance and as little as a 3% decrease in bodyweight from fluid loss can negatively impact strength [15].

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So to recap, a good place to initially set your fluid intake is per the bodyweight-based fluid recommendations shown below. While consuming this amount, assess how this intake level impacts the frequency and color of your urinations. If you urinate too frequently or if it is not very clear or infrequent, then you can further customize your fluid intake to your individual rates of fluid turnover by aiming for 5 clear urinations per day.

FLUID INTAKE GUIDELINES	
Based on Bodyweight (IMPERIAL)	2/3 of your bodyweight (lbs) in fluid ounces
Based on Bodyweight (METRIC)	1 L of fluids for every 23 kg of your bodyweight
Based on Urinations	5 clear urinations per day, 2 of those shortly after training

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Level 4

NUTRIENT TIMING AND FREQUENCY

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Now that we have covered overall caloric intake (Level 1), where those calories come from (Levels 2), and the micronutrients the foods we eat contain (Level 3), let's talk about how to distribute those calories over different periods of time. We will do this on both a broad and acute scale, diving into the timeline of the whole diet, over the course of a week, over the course of the day, and in relation to training.

Like all chapters thus far, these recommendations will be presented for both those dieting and gaining, as there are subtle differences between the two.

Diet Periodization (Diet Breaks)

Just like having high-volume blocks, high-intensity blocks and recovery blocks in your training program can help you achieve your goals more efficiently, we can arrange our diet to have higher-calorie periods and lower-calorie periods to achieve our goals more effectively as well. As popularized by The Guide to Flexible Dieting in recent years, the first method we'll discuss is a basic diet break [1].

The concept of a diet break is essentially just getting your oil changed before your car breaks down. Meaning, instead of dieting until you lose your mind and binge-eat, you simply implement a planned break to avoid the potential negative side effects that can come with a prolonged caloric deficit. Psychologically, this puts you in control of the process versus you breaking down as a consequence of the diet.

The original research on diet breaks found that this intentional "disruption" of weight control in fact did not disrupt weight loss. Those who implemented diet breaks were able to lose weight just as effectively as those who did not [2]. Additionally, time spent in a deficit can slowly but steadily reduce energy expenditure, thus making further weight loss more difficult as food has to be further reduced and energy expenditure increased. While much of this is unavoidable and simply a result of losing weight and thus burning less calories because you are moving less mass, a large component of this reduction in energy expenditure is what is coined "adaptive thermogenesis" [3] or what is known commonly as metabolic adaptation.

For those looking to get lean, this can be a headache. One study found that those who had recently lost 10% of their bodyweight or more had a total daily energy expenditure of 18% less on average than those at the same bodyweight who had not dieted [4]. Now this value of 18% is just the average value. Using the statistics presented in this study and extrapolating them to show the spread for 95% of the people studied, we'd get a range of 8-28%. What this means, is that some people near the end of their diet will be expending just over 90% of the calories that would be predicted by their bodyweight and activity, thus being only slightly affected by metabolic adaptation. On the other end of the spectrum, some would be nearly down to two thirds of the energy expenditure that would be expected [4]. As you can imagine, for some people "metabolic slowdown" can significantly frustrate their weight loss efforts.

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Fortunately, at least part of this adaptive component of energy expenditure reduction can be reversed by increasing calories [5]. Thus, a diet break can reverse some of this metabolic slowdown. It can also help to improve training quality and muscle mass retention by reducing fatigue and replenishing glycogen. But most importantly, a diet break allows you to take a mental time out, get your head right, and catch a breath of fresh air before you dive back under and go through another stint of your deficit.

Who Should Use A Diet Break?

I recommend this for people who are dieting for 3 months or longer and who have the time to implement this strategy without missing some sort of hard deadline. So unless you are behind schedule in terms of conditioning, a diet break can be awesome for contest prep athletes and I would highly recommend it.

Since it has shown to be so effective for all of the reasons listed in the previous section, I actually make it a point to plan for this in advance whenever possible, *and* to tell the client about it, as later down the line it is met with less resistance.

A good example of how you might set this up would be to take someone who has planned a 24 week contest preparation diet. In this scenario you might plan out 2 to 3 diet breaks, more or less evenly spread over the time frame. So for example, they might take week 8, 16, and then 24 (which would also be peak week, so it would marry up with carb loading) as diet breaks (*hopefully it's quite obvious that this dietary practice would not be necessary for someone in a lean gaining phase*).

How Do You Implement A Diet Break?

Essentially the goal is to get out of a calorie deficit and get close to maintenance. As a ballpark figure on how to do this, in most cases I recommend a 300-600 calorie increase to any days that are caloric-deficit days (depending on what maintenance was, body size, and how aggressive the diet is at the moment), concurrently with a drop of any cardio to about 50% of normal. The goal here is to eat as much as you can without gaining weight, or with only gaining minimal weight, as this weight will be predominantly water and glycogen. Diet breaks in general should last for 1 to 3 weeks.

What Might Happen During These Diet Breaks?

You could potentially gain, maintain or lose weight on a diet break. If you put on a little bit of weight, that's ok because you'll lose it right back once the break is over, and the weight gain very likely will only be the return of lost muscle tissue and glycogen anyway (see "Metabolic Magic?" from Level 1).

Starting (if you were stalled) or continuing to lose weight is also not uncommon, believe it or not. This is likely due to cortisol (a hormone that rises with stress and results in water retention) decreasing and water weight being lost, or energy expenditure increasing to the point where you are still in a deficit despite increasing calories and decreasing cardio, or likely a combination of these two factors.

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Lastly, you might more or less maintain weight yet simply feel and look better, be able to train harder, and get mentally and physically revitalized. This could be due to you being in a very small deficit or surplus, regaining lost lean mass and glycogen, while still losing fat or letting go of water retention.

Regardless of whether you gain, lose or maintain weight, any of these 3 outcomes are worth it, so I would highly recommend including diet breaks for anyone in an extended cutting phase. Doing so may land you on stage, or at the end of your cut, eating more food, doing less cardio, and retaining more muscle mass and sanity than if you hadn't implemented breaks.

Single-day Refeeds

On a slightly smaller scale, let's get into the manipulation of caloric and macronutrient spread across the week that might prove useful for strength and physique athletes. Some of these techniques are referred to as refeeds, or they may be referred to as carb cycling, calorie cycling or macro cycling. Regardless of the term, each means that you are distributing calories, and potentially macronutrients, in specific ways across days of the week in hopes of favorable outcomes.

What we will discuss is arguably the most common iteration, the periodic (often once per week) use of a 24 hour refeed where calories are brought to maintenance primarily via an increase in carbohydrate. While this has not yet been specifically studied, this strategy shares similarities with diet breaks in that one of its purposes is to attempt to reverse metabolic adaptation [6]. Not surprisingly, the increase in energy expenditure from 24 hours spent at or slightly above maintenance is small compared to what would be expected after a multi-week diet break. However, there may still be psychological and physiological benefits to 24 hour refeeds that make them worth implementing. The key is to ensure it is under your control, does not turn into a binge, and is a calculated increase to approximately maintenance calories via an increase predominantly in carbohydrate. In addition to potential psychological benefits, this strategy theoretically could help with glycogen replenishment, energy levels, and thus potentially performance and muscle mass retention over the course of a diet.

Despite these potential benefits, it is worth noting that the hormonal adaptations that occur from dieting may not be effectively reversed by a single day spent even far above maintenance calories [7]. It seems that not only is the magnitude of calories consumed during a refeed important, but the length of time spent out of a deficit as well. In fact, refeeds lasting 48 hours or longer at maintenance calories may be required to effectively begin reversing these hormonal adaptations [8].

However, this does not mean that 24 hour refeeds are without merit for the purpose of addressing metabolic adaptations while dieting. Even if not effective for *reversing* adaptations, spending a day out of a deficit is one less day that these adaptations are mounting and a refeed can act to reduce the cumulative effect of days spent dieting

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to some degree. Remember, just as it takes multiple days of refeeding to effectively reverse these adaptations, it takes multiple days of dieting for these adaptations to occur in the first place, and a refeed is a day of not dieting.

Who Should Use Single-day Refeeds?

I would recommend the use of 24 hour refeeds for most people when beginning a diet.

Metabolic adaptations are less of a concern when eating higher calories, doing less cardio, and when higher in body fat. Therefore, the issue of 24 hour refeeds not effectively reversing metabolic adaptation is not a concern when starting out. Additionally, the benefits of having a psychological break may improve adherence, and glycogen replenishment may aid performance and lean mass retention.

So How Do You Implement A Single-day Refeed?

The easiest way would be to raise calories to your calculated maintenance (as discussed in Level 1) for one day a week. Remember, that by doing so, this means you need to reduce your calories a bit more on the 6 dieting days to maintain the same weekly rate of weight loss.

For example, if you want to lose 1.5 lbs (~0.7 kg) per week, you need to produce a 5250 kcal deficit per week (3500 x 1.5). If you are dieting on 7 days, this is 750 calories per day that needs to be reduced. However, if you are dieting on 6 days and refeeding on 1, you need to reduce your 6 dieting days by 875 calories to reach the same weekly deficit. If you're not tracking macros, adjusting calories is all you have to do. If you are tracking macros, keep your protein and fat the same as your low days, and then drive your carbs up until you hit maintenance calories on your refeed.

What If You Are In A Gaining Phase?

You honestly probably won't benefit from cycling your macros or calories when not attempting to lose body fat. Although this method is unlikely to aid you in maximizing your physical results, it could possibly be used to improve your consistency if that's an issue.

Some people have an easy time following a diet Monday through Friday, but on the weekends they have a really tough time not going over their macro target. So if this describes your situation, just drop your calories on the week days a little bit and increase your targets for Saturday and Sunday to match your normal habits. You may find that doing this allows you to be more consistent with your weekly energy balance. Just don't go too crazy with caloric restriction on Monday through Friday, I would recommend going no more than 10% below maintenance calories.

Even a small reduction on 5 week days allows for a rather large increase on weekends. For example, if you were targeting a 200 kcal surplus per day during a gaining phase, and you simply ate at maintenance on Monday through Friday, this would allow you an additional 1000 kcals to consume on Saturday and Sunday on top of those days already

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being 200 kcals over maintenance!

Multi-day Refeeds & Carb Cycling

As I said, single day refeeds can prove to be useful for some reasons, but aren't the most effective tool for staving off metabolic adaptation. As you get leaner, deeper into your diet, and experience your body pushing back harder and harder, the potential utility of multi-day refeeds, or more frequent 24 hour refeeds increases.

Unlike weekly 24 hour refeeds, there is actually a fair bit of research on what could be described as multi-day refeeds. Most of these studies are on dieting strategies in which every other day is a dieting day, and every other day is at maintenance, or on studies where only 2 days per week are dieting days and the other 5 days are at or near maintenance.

One such study found that fat loss was higher in a group dieting only 2 days a week versus a group dieting continuously on all days, despite the same intended caloric restriction over time [9]. Also, in a review of studies using these strategies it was concluded that intermittent caloric restriction was superior for lean mass retention compared to restricting calories on all days [10].

So not only do multi-day refeeds have the potential for reversing metabolic adaptation to a degree, they can also result in a superior body composition at the end of your diet than if you had just dieted straight through.

Who Should Use Multi-day Refeeds?

Once you are below ~12% body fat if you are a male or ~20% body fat if you are a female, it would be a good idea to consider implementing multi-day refeeds. This is the point in a diet in which "the body fighting back" becomes increasingly more of an issue and proactive strategies for combating these problems are beneficial.

So How Do You Implement Multi-day Refeeds?

The easiest method is to simply have 2 refeed days in a row each week. You would implement this in the same exact way I explained for single day refeeds, except instead of spreading your deficit over 6 days, you would spread it over 5. Another way to implement multi-day refeeds is to have 3 days spread across the week as desired that are at maintenance. Once again, you would have only 4 days to spread the deficit over if you were to use this approach.

Are There Other Ways To Implement Refeeds?

There are for sure, and I'm not suggesting that 2 refeeds in a row, or 3 refeeds spread out over the week are the only or best ways to do it. Rather, these are just simple straightforward ways to do it. We don't have enough information to say what is optimal for most people, and even if we did, there would surely be a lot of individual variation.

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The general conclusions we can make from the available research is that there is a strong rationale for refeeding the leaner you get, and the leaner you get the more frequent refeeds should occur.

What About Having Higher Calorie Intakes On Training Days And Lower Intakes On Off Days?

If you get to the point in a diet where you are lean enough (~12% body fat or lower for men, 20% or lower for women) to start implementing multiple refeed days per week, there will be a larger disparity between your low days and high days. Using the same example as before but with 3 weekly refeeds, if the goal is to lose 1.5 lbs (~0.7 kg) per week (requiring a 5250 kcal deficit per week), your low days would be ~1300 kcal lower than your refeeds. This is a large deficit and could potentially impact training.

To offset this potential effect, it's not a bad idea to think about strategically placing your refeeds relative to your training days. However, placing them on the same day only makes sense if you train in the evening. Complete glycogen (your stored carbohydrate) resynthesis after exercise can take up to 24 hours [11]. So if you train in the morning or afternoon, you may not be getting the full benefits of a refeed if it is done on the same day that you train. So, if you are implementing three weekly refeeds, place them the day *before* your three highest-volume training days if you train in the afternoon or morning, or, place them on the same day if you train in the evening.

The premise of altering caloric intake on training versus non training days is likely unnecessary outside of the circumstances described above where you have a large energy disparity between low and refeed days and a high number of refeeds per week. The reason it is unnecessary, is that the difference in caloric expenditure on a training day versus an off day is quite small. Even a 90 minute, full body, resistance training routine where multiple sets of 10 reps with final set taken to failure burns 200-400 kcal on average depending on body size [12]. Meanwhile, if you hadn't gone to the gym and for that same hour and a half you sat on the couch you would have burned 100-150 calories. If instead you were doing light everyday activity, like grocery shopping in this period, you would have burned twice as much. So really, on a training day you are burning *at most,* a couple hundred calories more than on an off day. Thus, the premise that you have substantially different energy requirements on training versus non training days is flawed.

Finally, I would point out that adaptation from training does not occur only on the day of training; rather, adaptation is a continual process. So, while I don't have any evidence to cite, I would avoid setting calories too low on off days as well. While some have argued for this approach to limit fat gain in the offseason, it could also potentially limit muscle gain if it interfered with recovery. However, this would only be a concern with a large disparity in calorie intakes; small variations are unlikely to have any impact in my opinion.

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Meal Frequency

This simply refers to how many times you consume nutrients in a day. That might be the old-school recommendation of 7 to 8 meals per day from an IFBB professional bodybuilder, or that might be 2 to 3 meals per day in a certain window from someone in the intermittent fasting crowd. Generally, I'd say we should end up somewhere in the middle of those two extremes.

Now there is a decent amount of research on pretty much everything from 1 meal a day all the way up to 14 meals per day, and surprisingly limited investigation of the more moderate frequencies of 4 to 5 meals per day [13]. But collectively, what we can say is that if you are eating in the range of a relatively normal number of meals, say 3 to 6, this doesn't make a huge difference as far as the actual outcome of body composition [13, 14]. What we *can* say though is that some blips on the radar start to come up when you get on the extreme ends of very low or very high meal frequencies.

When you start touching the low end of 2 meals or less or the high end of more than 6 meals a day you can potentially start running into issues with adherence.

Less than 3 meals and you tend to go long periods of time without food and that can affect your hunger and ability to maintain consistency. You'll also eat these very large meals that can sometimes cause you to develop unhealthy relationships with food where you train yourself to consume enormous amount of calories at each sitting. Likewise, eating more than 6 meals a day you can run into the opposite problem. You're constantly focused on food because you're having these tiny little meals every hour or two that don't satisfy you. Either end of the spectrum can exacerbate hunger control [15, 16]

So my conservative recommendation that combines practical experience, theory, and the limited studies we have [13-20], is to consume somewhere between 3 to 6 meals per day in most cases. But that said, if you have previously found success with slightly more or less meals than this, that's not necessarily a bad thing. If you do fine on 2 meals per day or 7 meals per day, feel free to stick with it. There are plenty of high level competitive bodybuilders who have found a great deal of success following very high meal frequencies (6 to 8 meals per day), and likewise there are many folks who have followed an intermittent fasting approach to eating who have had personal or competitive success eating 2 meals per day (usually involving skipping breakfast) with only whey protein or BCAA prior to training.

In fact, when variations in meal frequency from 2 to 7 meals per day are studied no significant differences in energy expenditure are found [21, 22]. However, inconsistent meal timing across the week, as opposed to maintaining a consistent meal frequency, can decrease energy expenditure (to a small degree) [23] and insulin sensitivity [18]. So, if you are satiated by your current meal frequency setup, it is not socially stressful or inconvenient, and it works for you, don't think you have to change it.

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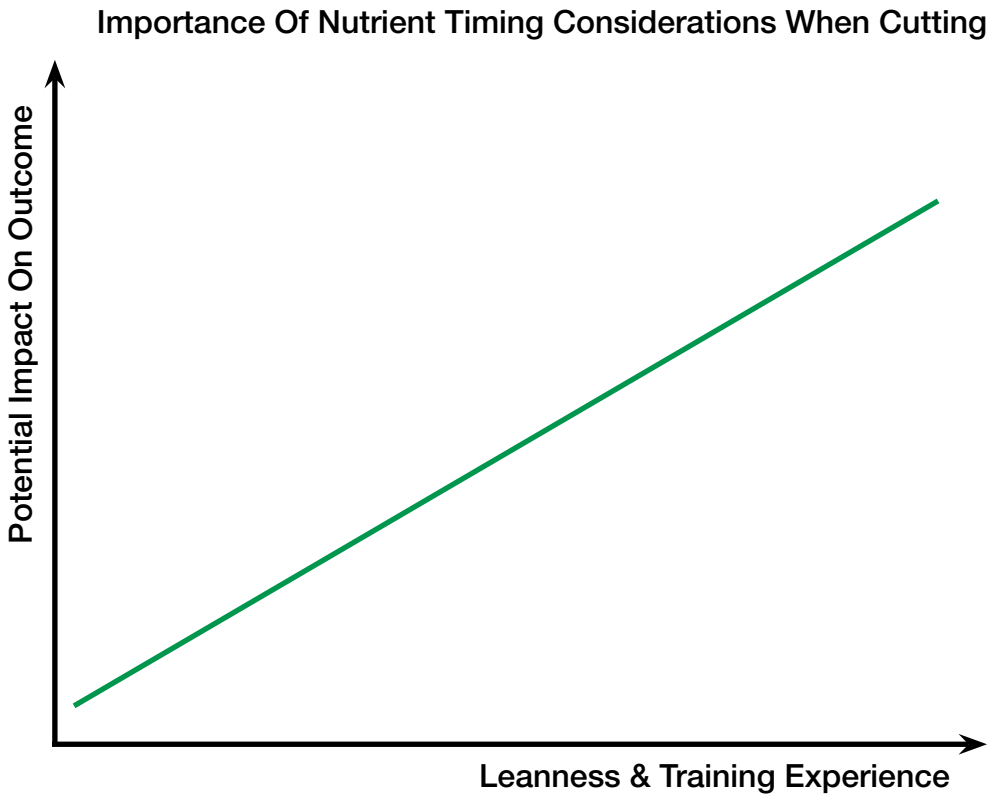
Believe it or not, I have a handful of clients who use an intermittent fasting approach as it helps them with adherence and satiety, and so does Andy. I also have some who eat a lot of meals per day. This is not black and white, and it is definitely not a deal breaker or maker in terms of your success. Rather, just recognize that most people will *probably* do well eating in the range of 3 to 6 meals per day, but not everyone.

Finally, for those of you who are larger individuals in a gaining phase with a high caloric intake, you might have trouble consuming the high number of calories that you need with a more moderate meal frequency. In your case, it might be better for you to aim for 6 or even 7 meals daily. Hunger is not going to be an issue that needs to be controlled. If anything, being too full can become an issue, so if you need to increase meal frequency to reduce meal size so you can get in 5000 calories, go for it.

Peri-workout Nutrition

This simply refers to the food you ingest before, during, and after your training, and it has been quite the controversial topic in our industry and the source of much confusion.

Before you go down the rabbit hole of peri-workout nutrition timing, please get this in your head:



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We all know people that have the minutiae of nutrient timing on perfect lock down before they have gotten their foundation steps in place. You see them scratching their heads 6 months down the line, looking no different now than they did before.

So, if you're a novice trainee (or coach them) you'll do well to focus on the bigger picture components of your training and nutrition plan before getting yourself wrapped up in the following details.

Got it? Good, then let's get into it!

There are plenty of arguments for eating carbohydrates immediately post-workout to maximize glycogen replenishment. There are also arguments to do this to spike insulin post workout for improved anabolism and growth. You can also find studies that recommend consuming carbohydrates pre-workout to enhance performance, and studies that suggest ingesting protein either pre or post-workout to speed up the build and repair process.

As you can tell, there are quite a few theories and statements but it is quite difficult to see how it all pertains to people interested in muscle and strength gains. So, let's go through all the different information and discuss what is relevant to the goals of muscle and strength gain and retention.

Post-workout Carbs

The primary argument for this is glycogen replenishment. However, in 90% of cases, bodybuilders and strength athletes don't need to worry about significant glycogen depletion. We're not doing multiple exhaustive bouts of glycogen depleting training for the same muscle group, and simply put, you will get full glycogen replenishment within a 24 hour period, even on a moderate carbohydrate diet.

If you were to do a regular weight training workout, let's say 9 to 12 sets for a body part, you might deplete those muscles up to as much as 40% of their glycogen levels [13]. After this, you won't train those muscles again for 2 to maybe 7 days depending on your training program. So you can see that glycogen resynthesis won't be an issue, because you will definitely replenish that body part's glycogen in the interim days before you train the muscle group a second time.

And for the second theory on consuming post-workout carbs for generating an insulin spike, protein actually spikes insulin on its own and to the same level as carbohydrate if consumed in an appropriate amount. More importantly, if you take all the studies that look at changes in muscle mass and strength between groups having either post-workout protein and carbohydrate or having that same protein and carbohydrate at some time not near training, only a minority of them show a superiority of post-workout consumption while the majority don't show a difference [24].

Therefore, post-workout carbohydrate intake is not quite as important as once thought in the context of someone just doing resistance training. That said, if you are doing

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cardio, especially anaerobic cardio like high-intensity interval training combined with resistance training in the same day, there may be a role.

Pre-workout Carbs

For pre-workout nutrition, the effects of consuming carbohydrate are mixed in the literature with some research suggesting carbohydrates have a role in aiding performance and others showing no benefit [25-32].

In practice, I have found that individual response is equally mixed. Some people seem to get a benefit consuming differing amounts of carbohydrate pre-workout for certain types of training and others do not. It tends to be very individual and very inconsistent, so play around with it.

If you eat too much too soon or near to your workout you're probably going to feel GI upset and may experience mild reactive hypoglycemia (low blood sugar) during training. If you don't eat for a *very* long period of time, especially if you miss a meal that you are used to having, you might get a little fatigued early, especially if you're doing a long workout.

Finally, whether you are dieting or not may play a role as well. A recently published study found that a group of lifters who consumed 28 g of carbohydrate prior to and after training, compared to a group that consumed 28 g of protein prior to and after training, had improved muscular endurance after 8 weeks of dieting [33]. That said, it's impossible to know whether simply having 56g of carbohydrate per training-day that the other group did not caused this difference in performance, or whether it was the timing of the carbohydrate. However, it may be a safe decision to simply ensure some amount of carbohydrate is consumed 1-2 hours pre workout in an easily digestible form while dieting.

Peri-workout Protein

When it comes to protein consumption, there is a decent argument that you should be consuming protein within some reasonable time frame around your training. However, what you've probably heard, that you should consume protein immediately post workout, is not as critical as we once believed. Remember it takes multiple hours to digest and absorb this macronutrient. So if you're eating 3 to 6 meals per day, chances are you're going to be digesting protein and have amino acids circulating in your body almost constantly. You're going to have the substrate to start the repair process available nearly at all times throughout the day.

So while the research does suggest that having protein around training versus not having protein around training may be of benefit [24], your normal eating pattern often ensures that protein is available to start the repair process anyway. Thus, in many cases you won't need to intentionally change your meal schedule or add a protein supplement to your diet to take advantage of a theoretical "anabolic window" (unless one trains fasted). In fact, a recent meta-analysis found that once you controlled for 24

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hour protein intake, the timing of protein within 1 hour of training had at best a very small effect on hypertrophy [34].

Far and away, consuming an appropriate amount of protein on a day-to-day basis rather than timing it specifically in relation to training has the largest effect influencing resistance training adaptation. However, a small effect is still an effect and from a pragmatic standpoint it is not at all difficult to consume some of your protein within 1-2 hours pre and post workout, and for this reason it has been recommended to consume 0.18 to 0.23 g/lb of bodyweight of protein (0.4 to 0.5 g/kg) 1-2 hours prior to and after performing resistance training to maximize muscular adaptations [24].

Considerations When Cutting

Now that we've covered all the theories, to make a safe recommendation that is appropriate for dieting individuals, I would say to eat a normal sized meal containing both carbohydrate and protein 1 to 2 hours before you train, and again 1 to 2 hours after you train. This meal should contain roughly 0.2 g per pound (0.4 to 0.5 g/kg) of protein, and should have roughly 10-20% of your daily carbohydrate intake (and 10-15% of your daily fat intake if you experience reactive hypoglycemia) depending on individual tolerance. This will ensure if you have to perform cardio and weight training on the same day that you will be able to perform as well as possible in both sessions, and will also hopefully prevent excessive fatigue when your food levels get low.

Considerations When Gaining

When you're gaining, you have a *much* bigger buffer in terms of glycogen storage, available food, and you aren't likely to be as fatigued from training. So, none of this matters as much. I would say that if you train no longer than 2.5 hours after having at least one meal containing protein in your body, and then eat another of your meals containing protein within 2 hours post training, you'll be all set. If anything, in some cases you may find because your food volume is so high, that you don't want to consume as large of a proportion of your food pre-training as you would while dieting to prevent GI distress.

Considerations For Extremely Active Athletes

Now there may be unique instances where you may be doing 2 or more continuous hours of training. If you're doing interval training right after your leg day, or if you are doing a really high-volume program and alternating between sets of push and pull, or if you're not a strength or physique athlete but do a lot of weight training right after your sport specific training, the timing of nutrition becomes more important. In these cases, it's not a bad idea to consider pre, post and maybe even intra-workout liquid nutrition to aid performance.

I would recommend consuming a drink that contains 8 to 15 grams of protein and 30-60 grams of carbohydrate mixed between pre- and intra-workout during training if it reaches or exceeds 2 hours of continuous, hard training [24]. That would be something

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like half a scoop of whey in Gatorade or something similar.

This is your typical endurance or team sport training intra-workout recommendation, but remember it's really *only* needed if you're doing 2 hours plus of continuous hard training. Not 1 hour and 45 minutes of just weights with most of that being rest periods.

I know some of you are thinking "yeah I lift for 2 hours or longer quite frequently". Even then these recommendations don't apply, because I literally mean more than 2 hours of *continuous* training with minimal rest, which to be honest, if you are doing as a bodybuilder or a powerlifter...you probably shouldn't be.

SUMMARY OF NUTRIENT TIMING RECOMMENDATIONS				
	DIET BREAKS	**REFEEDS**	**MEAL FREQUENCY**	**PERI-WORKOUT**
GAINING	N/A	N/A	3-6/day, More if needed to reach daily kcal goal	0.18-0.23 g/lb of protein (0.4-0.5 g/kg) 1-2 hr pre & post workout
CUTTING	~1-2 wk diet break for every ~8 wks of dieting	Increasing from 1-3 refeeds/wk as you get leaner	3-6/day, Greater or fewer meals if more satiating	10-20% carb intake pre, 0.18-0.23 g/lb of protein (0.4-0.5 g/kg) 1-2 hr pre & post workout

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Level 5 SUPPLEMENTATION

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Dietary supplements are substances produced with the intentions of providing nutrients to your body that it would not otherwise get from your current diet. Sometimes they are ingested to achieve a recommended healthy baseline of certain nutrients, and other times they are used to reach higher levels of certain nutrients than could not be normally attained by most diets to provide additional benefits.

These substances are often derived from components of typical food sources that you already consume or are made from naturally occurring plant products. They are specifically manufactured with the goals of benefiting your health, performance, or recovery through various mechanisms.

Supplements are the least important part of The Pyramid and are not necessary for success, barring that you do not have some type of nutritional deficiency or condition that requires them. In fact, in the grand scheme of things, the vast majority of supplements on the market don't do anything except burn a hole in your wallet. That said, *some* are based on sound science and are therefore worth discussing as they can provide a small benefit. And because there are so many ineffective or unnecessary supplements on the market, we also need to discuss how to differentiate between the useful ones and the not so useful ones. Once we've learned how to evaluate them, I will present a list of recommendations and doses for you to consider.

So to start, let's dive into the three factors that must absolutely be considered when analyzing any nutritional supplement: quality, validity, and effectiveness.

Protecting Yourself & Ensuring Quality

Quality of the product refers to whether or not the ingredient-claims on the label are actually true. Does the bottle actually contain what the label says it does? And how can we be sure of this? Now before I actually get into ways to ensure quality, let me explain *why* this is such an important aspect of supplementation to consider.

The majority of products in the bodybuilding industry are propped up by sexy marketing and athletes paid to endorse them. At best, there is usually a bit of pseudoscience mixed in. At worst, a surprising number of products simply don't contain the quality or the amount of the active ingredients that their label claims [1-3] and some actually have ingredients in them that should not be there at all or that are either unsafe, illegal or banned, or both [4-6].

If you are a recreational lifter, it is not outside the realm of possibility that if you aren't well informed and careful, you could end up taking something for the purpose of making your healthier, which is actually harmful.

If you are a competitive natural athlete in an organization that expects you to be drug free and regularly drug tests you, not only could your health be negatively affected, but you could in fact inadvertently end your career if you were to unknowingly consume a contaminated supplement [7].

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Let's talk about how you can protect yourself and ensure that what you think you are taking is what you are actually taking.

Laboratory Analysis

One of the ways to check the quality is to make sure the product has been run through a lab analysis. This information, which is sometimes on the product label and most often on the company's website, will include some type of report showing that a third party lab analysis found the contents of the bottle to be in line with the claims made by the product creator. Basically, some other lab verifies that what the supplement company claims is in their product, *is* actually in it, and nothing else.

Unfortunately, this is an incredibly rare practice in the supplement industry. Yet despite how rarely this is done, I still would recommend sticking to supplements that have been validated by an external lab analysis (especially if you are a drug-tested athlete). Believe it or not, there are actually some supplement companies that send lab reports to their customers with their orders. This is a mark of a quality company.

Another quality check would be a third party certification. This could be something like the U.S. Pharmacopeial Convention (USP), NSF International (NSF), or Good Manufacturing Practices (GMP) stamps that you'll see on bottles. These certifying stamps indicate that there is a third-party company without any biased interest in that supplement's success that has come into the facility and approved the operations and practices that are involved in that product's creation. So supplements that have USP, NSF, or GMP certifications are generally more likely to have some basic level of quality and to meet the claims listed on their labels.

Similarly, the use of reliable suppliers can help ensure the quality of a product. For example, CreaPure is a company that produces, and has a stake interest in being a high-grade supplier of creatine monohydrate. Other supplement companies will often license the use of CreaPure in the supplements they sell because of the ensured quality. So, for example if you find some product that combines a few compounds into a pre-workout supplement, one of them being CreaPure, you can ensure that at least the quality of the creatine monohydrate will be satisfactory.

Proprietary Blends

I would also recommend *not* getting products with proprietary blends. A proprietary blend is where a company trademarks a combination of substances that they mix together and then gives that mixture a name of their choosing. They are required to list the ingredients, but not the dosages. For example, a label might say it includes 30 grams of a "muscle blend" and lists all the ingredients, but doesn't tell you the individual amount of each ingredient.

This is basically a way to say "look, this is our special sauce, we don't want people to reproduce it, and so we're not going to tell you exactly what's in it." Too often, this becomes a way to disguise the fact that the individual ingredients are too under dosed

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to be effective, and to get you to focus on the fact that it has a lot of ingredients and to think "therefore it must work". The use of this practice to fool consumers is why I recommend avoiding proprietary blends altogether.

So in my opinion, before purchasing a supplement, you should ensure the majority of the above factors are in place to increase the likelihood that you are buying a product of high overall quality. If you can't ensure all of the above conditions are met, at least ensure some combination of these factors that you are comfortable with is met before wasting your time or money on false claims, lower dosages than expected, or even potentially tainted products.

Protein Powder Scams

Here's a good example of the industry's shenanigans combining both the factors above with one of the most commonly used supplements, protein powder.

We bros like protein powder. It's a quick, convenient, and cost effective way to hit our daily protein targets. Whey protein is not the cheapest, but it is popular due to the high BCAA content, particularly leucine, which is critical to the muscle building process.

Now, with consumers becoming wiser there is a rising demand for products that claim to have been lab tested, but this comes at a time of overall rising global demand (and thus prices). With consumers becoming sensitive to these price increases, and a lack of general education about what they should be looking for on the packet, the incentives for companies to cut costs by cheating the system are all there, and many do.

I'm talking about the rise of the phenomenon known as 'protein spiking'.

The way it works is this: some labs test for the total amino acid content rather than the amounts of the individual amino acids themselves. This means that protein companies can dump cheap amino acids into the mix (mainly glycine and taurine), skimping on the actual whey content, which is expensive, and yet still pass some quality tests.

Here are some red flags to look out for when choosing a whey powder:

1. The cost per pound / kilo of claimed protein content is considerably cheaper than average. Whey is a commodity traded on the open market. You can be ripped off and pay way too much (Andy has seen 11x market price protein from one "luxury brand" gym in Japan), but you won't ever find it significantly cheaper than everything else without there being something dodgy going on.

2. It has a proprietary blend (or doesn't list leucine content).

3. Leucine content, when listed, is lower than 2.7 g per 25 g of protein content (the BCAA content of whey is 25%, leucine should be 11%).

If your protein powder doesn't pass those checks, you're rolling the dice with the quality of what you're getting.

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Supplement Validity & Effectiveness

Although I will group validity and effectiveness together in discussion, they are not exactly the same thing. The question "is this supplement valid" does not always tell you if it is effective for your specific goals. Really, the question boils down to "does this supplement benefit *you*?" To determine if a supplement benefits you, you have to determine not only if it has an effect, but also if it is an effect that is relevant to your goals, and if the effect is significant enough to have any measurable outcome.

We have to keep in mind that some supplements actually have an effect, but it's just so insignificant that it's not worthwhile to purchase. Other supplements have an effect, but the effects themselves are irrelevant to the goals of building muscle or strength. Luckily, there are a few great places on the internet that can help us with finding out this information.

In my opinion, Examine.com is probably *the* best website about supplements that exists in the world today. This site is a collection of human research on almost every nutritional supplement in existence. When you arrive at the site, you can do a simple search to bring up a full Wikipedia-type page on nearly every supplement available, and it includes information on usage, dosage, chemical compound structure, side effects, references, etc.

Another feature is what they've coined the "Human Effect Matrix", which is a table based on the available human research on a supplement. It includes the studied effect or outcome, how the supplement impacts this outcome (increases or decreases it), the magnitude of the change, what percentage of the studies on this outcome share the same finding, and a ranking for the research quality. Essentially, you can use this website to help you make decisions about which supplements to purchase by answering questions like, "Does this increase or decrease a given variable?", "Is it a small, moderate or large effect?", and "How much agreement does there appear to be in the research?"

You could use this site to look up something like creatine monohydrate, versus tribulus terrestris; both of these substances are often recommended in bodybuilding communities. But what you'll find is that one of these has a whole lot of scientific backing (and you can tell exactly how and exactly why), while the other substance, not so much.

Another website I'd recommend is the Journal of the International Society of Sports Nutrition (JISSN). This is an open-access online journal that does a lot of sports nutrition research and has position stances on things like protein intake, HMB, creatine use in adolescents, nutrient timing, and many other pressing topics in the sports nutrition and supplement industry. While the JISSN doesn't spoon feed you the info on a silver platter, being open-access it does allow those who want to dig deeper and learn the science and mechanisms behind the recommendations to really dig in and research the supplements in question.

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In essence, between those two sites, Examine.com and JISSN.com, you'll be able to easily check up on the validity and effectiveness of any supplement, which will in turn help you realize whether or not it's worthwhile for you and your goals.

Be Cautious With New Supplements

As a final note before moving onto the recommended supplements, a word of caution on purchasing new supplements that are released with only minimal scientific support for their use. New supplements come out all the time; most are not supported by any studies whatsoever. Less commonly, some supplements that come out are supported poorly by an animal study, short-term study or an in vitro study that doesn't directly measure fat loss, muscle gain or performance, but measures an outcome that is *related* to fat loss, muscle gain or performance. Even more rarely, a new supplement will come out that actually does have a scientific study that supports its effectiveness directly.

- In the first case where a supplement has no supporting evidence, simply ignore it.
- In the second example where the evidence is shaky at best, you can keep an eye on the supplement, but don't get your hopes up. If it really was effective they would have cited studies that demonstrated its effects in an unambiguous way to show its effectiveness.
- In the final case where a new supplement comes out with direct scientific support, my advice is to wait until a second group of researchers has confirmed the findings of the first group to ensure that the findings are legit.

Researcher bias, small sample sizes, and dare I say it financial influence from the industry can all potentially influence study outcomes. The smart thing to do would be to wait until there is a preponderance of evidence supporting the use of a supplement before you decide to take it.

Think I'm being overly cautious?

Let me give you a specific example. Remember D-Aspartic acid? A study came out in 2009 that found it increased testosterone production in men to a rather impressive degree [8]. Immediately, supplement companies jumped on the band wagon and started selling this commercially as a muscle builder, libido enhancer and performance enhancer (despite the fact that the study did not measure changes in muscle mass, performance or libido). Over and over I cautioned for people to wait until further studies had been performed before taking the supplement, considering this was the one and only human trial. Many ignored me, and then to their surprise (certainly not to my surprise) a study came out in 2013 that found D-Aspartic acid not only failed to increase testosterone levels, but most importantly it had no effect on body composition or strength [9]. Once again, another supplement proves to just be a waste of money right? Wrong. A study recently came out that actually found D-Aspartic acid *decreased* testosterone levels in resistance-trained men [10]! So not only did this compound end up being a waste of money, it was found to potentially be counterproductive to the goals of those taking it.

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Even the most effective supplements that have been tried and tested time and time again for decades that can help you reach your goals, make only a very small impact. Likely, the impact they will make might not even be discernible to you. So, don't get caught up in marketing hype of some new dodgy supplement and think that by waiting a year or two until more research comes out that you'd be missing out on potential gains by not taking it. In fact, jumping on a bandwagon too early could be much worse of an idea. Remember, 1000's of supplements have come out over the years, and 99.9% of them have disappeared because they proved useless. Assuming that a new supplement is the "next big thing" without sufficient proof is illogical and is just buying into marketing.

Recommended Supplement List

Keeping in mind how adamant I am regarding the combined quality, validity, and effectiveness of any substance, I'd like to present a compiled list of the few that I personally think might be useful for those of us interested in gaining muscle and strength.

Now there's always some subjectivity in deciding whether these supplements are worth being on this list or not, and as with all of my advice, feel free to disagree with it based on your own judgement. Additionally, it is quite likely that this list will prove to be outdated in a few years as more research comes out.

That said, we're going to introduce these substances and discuss why they're worthwhile, when they're worthwhile, and what type of doses you should be consuming based on the research relevant to our population (people who want to get bigger, stronger or leaner).

Multivitamins

The common bodybuilding multivitamin is very high in your water-soluble vitamins, and probably even some of your fat-soluble ones, to the amount of seeing up to 4 digit percentages of the recommended daily allowance (RDA). Something like 4000% of a certain vitamin is not that uncommon for some products in supplement stores.

This gross and unnecessary overdose is exactly why I don't really like those massive bodybuilding vitamins that are either giant horse pills or a large packet of pills that you're supposed to take on a daily basis. If you've got the first 3 levels of The Pyramid under control and in order, you should honestly only need a regular multivitamin once a day that covers your bases and protects against deficiencies. You can also find many basic multivitamins that are geared toward common deficiencies for a certain population, like those specifically made for women with emphases on calcium, magnesium, zinc and iron.

Now that being said, in general, multivitamin use appears to be not harmful, and may confer a small protective health benefit to long-term users [11]. Additionally, logic would tell you that when you are eating less food, you will also be eating less micronutrients

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and some studies of popular diets have found that they are actually micronutrient deficient [12]. Therefore, I would say that during a cut it's probably a good idea to create a nutritional "safety net" and have a daily multivitamin that's of high quality but that isn't overdosed. If you're in a lean gaining phase, this is most likely not necessary, but might be helpful. You may be one of those people who benefits from multivitamins year round if you know you don't have a whole lot of variety in your food source selection or if you know you're not so good with the fruit and vegetable intake guidelines. However, a multivitamin should not be seen as an excuse to eat a micronutrient poor diet and for those who don't consume adequate micronutrients, a better solution than trying to make up for this with supplementation would be to work towards consuming a more balanced diet.

Essential Fatty Acids - (EPA & DHA)

Of the essential fatty acids (EFA's), eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), which typically come from fish oil supplementation, have been found to have a host of potential health benefits [13]. If you don't eat fish or don't like taking fish oil, you can also get EPA and DHA from an algae supplement, which is what the fish eat that gives them the EPA and DHA that we are looking for.

When appropriately dosed, EFA's help with leptin signaling in the brain, reducing inflammation, enhancing mood, and reducing disease factor risk. They can also aid in joint recovery and have shown potential for some metabolic benefits as well.

Both while cutting or lean gaining, I would recommend getting enough EPA and DHA (combined) to fall within the 2 to 3 gram range per day. To check this, look at the back of your fish oil (or algae oil) container, add together the EPA and DHA, and look at the serving size. Perhaps the label says that there are 400 mg per serving of EPA and DHA when combined. This would mean that you need 5 servings to get 2 grams (There are 1000 mg per gram).

Vitamin D3

Vitamin D is primarily produced in our body as a result of direct contact with sunlight. Having insufficient levels of vitamin D in the body can compromise the immune system [14], which can be a disaster for someone who is training hard, dieting, or attempting to perform any type of activity at a high level. Vitamin D is also a precursor to hormones, and low levels have been linked potentially to seasonal depression and other mood alterations [15].

Vitamin D deficiency rates are a lot higher than we once thought, and being deficient in vitamin D can potentially negatively impact muscular performance, immune function, and hormonal status [16]. Thus, it's good idea to supplement accordingly if you don't get much direct sunlight, have dark skin, or a combination thereof. A basic dosing recommendation would be to take anywhere from 9-36 IU/lb/day (20-80 IU/kg/day) of vitamin D3 based on sunlight exposure. For those who find supplements that don't list

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the amount of Vitamin D3 in IU's, the equivalent dose in micrograms is 0.225 to 0.900 mcg/lbs/day (0.5 to 2 mcg/kg/day).

So if you work or train outdoors on a daily basis, you might not benefit from supplemental vitamin D3 at all. Perhaps taking 9 IU/lb (20 IU/kg) at most to be extra safe that you're getting enough would be a good idea. If you are someone on the opposite extreme end of the sunlight exposure spectrum, like me, it might be more appropriate to take the full 36 IU/lb (80 IU/kg). I typically sit in front of a computer all day, go downstairs and train in an Olympic lifting facility that's indoors, and then get immediately in a car to drive home and sit in front of a computer again (unfortunately sunlight through a window doesn't provide you vitamin D [17]). However, in the summer I do get outside a fair bit. So, I regularly make sure to supplement the maximal effective dose throughout the winter, and in the summer months I often don't take it at all.

If possible, the absolute best route would be to get your bloodwork checked to see where your levels are and to see if you're deficient. Otherwise, just use your best estimate in the range provided based on lifestyle and exposure to sunlight.

Creatine Monohydrate

Now that our basic daily supplements are covered, let's dive into the performance supplements that made this list of recommendations.

Creatine monohydrate is by far the most tried and true, most affordable, and most effective of all the creatine variants [18]. It is the original, and many subsequent variants of creatine are either inferior [19], or cost more without giving any additional benefit [20]. So every time I refer to creatine, creatine monohydrate is the specific type that I am referring to.

Creatine is produced in the body and so is already present. We get it from animal products, mostly in meat, and it is more abundant in raw meat. When meat is cooked it degrades the creatine content [21], which is why it is difficult to get the performance-enhancing benefits without consuming this as a supplement.

To get creatine stores up to levels where they can benefit strength, power production, muscle fullness, and ultimately your long-term ability to produce more muscle mass over time, I would recommend ingesting 0.018 g/lb of bodyweight per day (0.04 g/kg/day). It will take a couple weeks of ingesting this amount per day to reach supplemental creatine levels, but after that point you can just maintain those levels by continuing to take the dose, like topping off your gas tank.

And last but not least, it's important to note that for long-term consumption, timing doesn't matter. It doesn't need to be taken with carbs, it doesn't need to be loaded, it doesn't need to be taken pre-workout, and it doesn't need to be taken post workout. All the benefits associated with creatine timing, whether it's taken with carbs, or if creatine is loaded in large amounts, are strictly related to the first couple weeks of consumption where the goal is to get to supplemental levels. It has nothing to do with long-term use

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and whether it takes you 5 days or 21 days to reach supplemental levels of creatine has a less than negligible effect on long-term gains. So, just take the daily dose I recommend and you'll be all set to receive its benefits.

Caffeine

Caffeine is the only pre-workout supplement that I would recommend, if anything is taken at all. Other pre-workout supplements are just variants of caffeine, or are just piggy-backed with caffeine. There are some other pre-workout supplements that might be effective, but I'd stick to caffeine as there is a multitude of research behind its effectiveness in enhancing performance and suppressing tiredness while other stimulants don't have nearly as much behind them.

Instead of giving a blanket recommendation for everyone, or different ones for cutting and bulking, I'm going to break this into two different dosages based on what effect you are trying to achieve, and also considering that you develop tolerance to caffeine and some of its effects with chronic usage.

To suppress tiredness doses of 0.45-1.40 mg/lb/day (1-3 mg/kg/day) are appropriate. Fortunately, the suppression of tiredness does not appear to be effected by caffeine tolerance [22]. For acute resistance-training performance enhancing effects, take 1.81-2.72 mg/lb (4-6 mg/kg) approximately 30 min before training no more than 2 times per week.

The reason I stress to take caffeine so infrequently if your goal is performance enhancement, is because the more you take it, the more resistant you get to the performance enhancing effects [23]. We must remember that it is a stimulant of the nervous system and you can become tolerant very quickly.

So if you feel great and you're ready to train, you don't need to take a pre-workout at all. But if you are feeling down and are anticipating a terrible workout, then it might be a good day to maybe take some. It's a valuable strategy to keep your caffeine intake as low as possible in order to allow yourself the ability to get those stimulatory effects when it is truly needed. Also, think about the times where suppressing tiredness might be more beneficial than enhancing performance. If you are on an extended, rather aggressive diet, it may be more beneficial for *overall* performance to take a tiredness-suppression dose regularly to maintain performance on all days, than it would be to take a performance enhancing dose once or twice a week.

Finally, even though the above recommendations are given for maximal effectiveness, I'd recommend starting with half of these dosages for either suppressing tiredness or enhancing performance because of individual response. If you are not used to stimulants, or if you take too much you may feel jittery and unfocused, which can be an equally undesirable effect to feeling tired or low in energy. To start out, take the minimum effective dose to get the effect you desire.

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Beta-Alanine

With all of the aforementioned "top tier" health and performance supplements taken care of, let's dive into the rest of the list which includes supplements to be taken situationally. These are by no means necessary, but they can potentially be helpful in certain instances.

Beta-alanine can be thought of as the muscle endurance version of creatine monohydrate. If you think of creatine for power, think of beta-alanine for longer anaerobic performances.

I consider beta alanine as a conditionally beneficial supplement because it only has a small performance enhancing effect when efforts reach a duration of at least 60 seconds [24]. So, for powerlifters, Olympic lifters, and anyone training primarily for strength, it doesn't serve a purpose. With the typical tempo that most people lift with, you'd need to be doing sets of 15 reps+ to get a benefit from beta alanine. So perhaps bodybuilders could benefit from taking beta alanine, but only if they are doing a large volume of high-rep work. However, there is not a requisite rep range for hypertrophy training per se. Equal hypertrophy can be achieved with both 10RM and 3RM loads given equated volume; the only advantage of moderate repetition ranges (8-12) is that it is easier to accumulate volume compared to using heavier loads which requires more sets, long rest periods and thus takes much longer [25].

Simply put, to optimize hypertrophy enough volume needs to be performed with loads that are "heavy enough", because high-rep, low-load sets (20RM+) don't produce as much muscle growth as moderate and heavy loads when volume is equated [26]. In fact, to get comparable muscle growth to that which can be achieved with heavier loads, one may have to perform three times the volume with high-repetition, low-load training [27]. Therefore, I would argue that only during phases of training where high reps (15+) are emphasized, or potentially when one is performing HIIT cardio using intervals lasting 60 seconds or longer, would it be advisable to consider beta alanine supplementation.

If you are regularly performing continuous efforts lasting 60 seconds or longer, beta alanine might be worth considering, and taking 3-4g/day can result in a potential small benefit to you.

Branched Chain Amino Acids (BCAAs)

Whether you are cutting or lean gaining, I would have to say that BCAAs probably aren't going to provide a benefit to you if your protein intake is already high enough. A high protein diet (such as recommended in this book) will provide you with ample amounts of BCAAs.

Despite this fact, BCAAs are one of the most frequently used supplements for bodybuilding [28], which is odd considering there is only one study directly relevant to muscle and strength development that supports their use [29]. This popularity is even less founded considering this study was a poster presentation rather than a peer

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reviewed journal article, and considering that dietary controls weren't put in place to account for total daily protein intake (which is important since a high protein diet contains a high content of BCAAs). Outside of this one poster presentation that should be interpreted with caution, there is scant evidence to support BCAA use.

A recent paper came out that found handball athletes had a reduction in fatigue while taking a combined arginine and BCAA supplement compared to placebo during a multi-day simulated competitive event [30]. However, it is unclear whether this effect can be isolated to BCAAs (rather than arginine or the two combined) and the fatigue reducing effects may only manifest during multi-day efforts considering a similar study on wrestlers found no effect during a single protracted competitive effort [31].

In terms of body composition, an older study found a greater reduction in abdominal fat in dieting wrestlers who replaced the vast majority of their daily protein intake with BCAAs compared to another group of wrestlers who replaced the vast majority of their daily protein intake with soy protein [32]. However, neither condition was representative of a normal diet. The extreme intakes of both soy and BCAAs in this experiment make it difficult to draw meaningful conclusions from this study.

Finally, there is evidence that aerobic exercise performance can be improved when consuming isocaloric amounts of BCAAs compared to carbohydrate but only when glycogen depleted [33]. When not glycogen depleted, BCAAs may reduce perceived exertion, but are not as effective at aiding performance as an isocaloric amount of carbohydrate [34].

So, the take home here is that if you are consuming a diet adequate in carbohydrates, protein, and total calories, you are unlikely to get any benefit from supplemental BCAAs, especially if you are hoping they will improve resistance training performance. On the other hand, if you are performing cardio and have to do so fasted or while consuming a low-calorie or low-carbohydrate diet (which could potentially result in glycogen depletion), you might potentially benefit from BCAA supplementation. However, this is a very narrow rationale for BCAA supplementation indeed.

So, if you are glycogen-depleted, on low carbs, or training fasted, it may benefit you to have a dose of 0.09 g/lb (0.2 g/kg) pre-workout as it helps delay time to fatigue. For example, let's say you were going to do your high intensity interval training (HIIT) right after a leg day while dieting on low carbs. It might be beneficial to take BCAA while you're recovering after squats and you're on your way to do your HIIT.

So, while the reasons to take BCAA are relatively obscure, there still could potentially be a situation where it would be worth considering. As a final note, as discussed in Level 4, consuming protein in the amount of 0.18-0.23 g/lb (0.4-0.5 g/kg) pre-workout if taken as whey protein could elicit similar effects, so long as digestion is not an issue.

Beta-Hydroxy-Beta-Methylbutyrate (HMB):

HMB is a metabolite of leucine that has been shown to act as an anti-catabolic in

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populations struggling with muscle wasting diseases or conditions [35]. However, effects on performance and body composition, especially in trained populations are at best mixed [36, 37]. While it has not been studied directly, there is a theoretical rationale that HMB could potentially aid muscle maintenance while dieting [28] but if this is the case, the benefit would likely be small if it even occurred.

Because HMB is not as highly researched and validated as some other supplements at the top of this list, consider HMB as something that *may* be effective if you have the money lying around and if you are doing a relatively hardcore diet...but don't expect much.

So if you are cutting and have the financial means, I would recommend ingesting 3 grams of HMB about 30 minutes before training sessions. This dose might also be beneficial for people who are in an intense overreaching cycle or are acutely increasing their volume for whatever reason. But regardless of your situation, if you are in a gaining phase and have plenty of food, you definitely don't need to be taking this supplement.

Again, this is very low on my list of importance and should only be considered if you really have the money just lying around and fall into the category of being at risk of muscle loss based on long dietary cuts or extreme overreaching.

SUPPLEMENT	DOSING RECOMMENDATIONS
MULTIVITAMIN	Cutting – 1 per day normal dosed multi; Gaining – not normally needed
EFA'S	2-3 g/day EPA and DHA combined
VITAMIN D3	9-36 IU/lb/day (20-80 IU/kg/day) based on sun exposure
CREATINE	0.018 g/lb/day (0.04 g/kg/day)
CAFFEINE	Suppress tiredness – 0.45-1.40 mg/lb/day (1-3 mg/kg/day); Pre-Workout to enhance performance – 1.81-2.72 mg/lb (4-6 mg/kg) 1-2x/week max
BETA-ALANINE	3-4 g/day only if doing continuous high intensity exercise for 60 sec+
BCAA	0.09 g/lb (0.2 g/kg) pre-workout if glycogen depleted, low carb, or fasted
HMB	Cutting – 3g/day pre workout if extra money available; Gaining – Not needed

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BEHAVIOR & LIFESTYLE

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Now that we are aware of all Pyramid Levels, their order of importance, and how to give ourselves guidelines in alignment with them, how do we put all of that into practice? Although this can be a loaded question, it is definitely the most important one of this entire guide.

Because what good is a nutrition plan if you cannot follow it? And unless you're in a contest prep fat-loss phase, what good is a nutrition plan if it is not maintainable in the long run? These are some of the queries and topics that I will address here in this final portion of our guide.

The bulk of this section will go over a tier-based system of nutritional tracking methods that you can use based on how strict you need to be depending on your situation. I'll also give some general guidelines on how to eat out with loved ones, and even consume some alcohol in moderation if you so choose.

Tracking Your Numbers With The 3-Tiered System

Throughout this text thus far I've made it a point to always acknowledge that contest prep is different than a fat-loss phase, which is different than maintenance, which is different than off-season gaining. This very broad spectrum of goals and seasons throughout the life of an athlete is not often addressed in nutritional programming, which is why many individuals get stuck in social isolation and can easily adopt those dreaded "all or nothing" ways of thinking.

Without context, everything appears to be of equal importance at all times and a great deal of effort can be spent micromanaging a laundry list of dietary details. So the question becomes, which of the details needs to be managed and when? While The Nutrition Pyramid in and of itself can help you navigate this question, it is also true that some elements of The Pyramid can become nearly irrelevant in certain circumstances.

An analogy I often use to describe the different nutritional needs of an offseason bodybuilder or powerlifter compared to a bodybuilder during contest prep or a powerlifter dieting to a lighter weight class is that the former is walking a wide path while the latter is walking an ever narrowing path slowly becoming a tight rope.

What I mean by this is that when one is lean and calorically restricted, they have less leeway in their nutrition before it has the potential to negatively impact them. A greater amount of energy is created by metabolizing body protein in lean individuals compared to those overweight [1], and when lean individuals are calorie restricted they experience a reduction in testosterone [2] while those who are overweight likely will not [3]. For these and other reasons, the leaner you are the more likely you are to lose lean body mass during weight loss [4].

Additionally, resistance training depletes glycogen to a degree [5] and when glycogen depleted (which is more likely to occur while dieting), muscular performance can degrade [6]. Resistance training is also partially fueled by intra-muscular triglyceride

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which is depleted to a degree alongside glycogen during resistance training [7] and a diet low in fat may not completely replenish intra-muscular triglyceride levels [8]. Since both carbohydrate and fat will likely be restricted while dieting to reach the requisite caloric deficit, performance is more likely to be negatively affected while dieting. Logically, the magnitude of the caloric deficit has a direct impact on how much dietary fat and carbohydrate is consumed and thus it is no wonder that faster rates of weight loss (achieved through larger caloric deficits) can result in poorer strength performance [9] and muscle maintenance compared to slower rates [10].

In addition to these physiological impacts, the psychological stress associated with intensive weight loss can be much higher than the stress experienced during a non-energy restricted period [11, 12].

Matching Precision To Needs

Due to all of the above, there are disparate needs based on nutritional status. I hope I have conveyed thoroughly that contest-prep dieters require high levels of consistency and accuracy, which usually comes with far less flexibility. However, when one isn't dieting and is in the "off-season" or simply doesn't need to diet (such as in the case of a powerlifter staying in the same weight class), what needs should be focused on and what degree of precision and accuracy are required on a day to day basis? Well, this comes down to interpreting concepts and theories to create a framework for practical application.

Nutritional science and exercise science rarely tell us explicitly what to do. Rather, they help us determine what should be measured and managed. We then have to translate that knowledge into something actionable for us as athletes.

An example of this would be managing macronutrient ranges and the size of a caloric deficit. When coaching my contest-prep athletes, I typically give them a daily target for the three macronutrients and decrease or increase them based on the rate of weight loss that occurs on a week to week basis (while also looking at body comp change via pictures and weight room performance). During weight loss, the specific balance of macronutrients becomes more important because there is an imposed maximum energy intake and an increase or decrease in any macronutrient necessitates the subsequent increase or decrease in another to maintain the caloric deficit.

For this reason, I use a relatively tight range for the macronutrients, often + or - 5 to 10 g while dieting. This degree of accuracy helps the dieting athlete walk the "tight rope" as described in my analogy above. However, it also requires attention to detail, the use of tracking software or applications, a digital food scale, abstaining from eating out the majority of the time, and planning some meals or even entire days in advance when traveling. This degree of attention to detail is certainly sustainable for finite time periods and is the cornerstone of success for all of my clients with shorter-term goals requiring calorie restriction.

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On the other hand, when we're talking long-term success and sustainability as an athlete, adherence is the most important variable. And in my experience as an athlete and coach, the harder it is to stick to a plan, the less likely it is that you are going to achieve your goals using it.

So in a culture where "if it fits your macros" or "follow a meal plan and eat clean all the time" are often presented as the only two ways of doing things, it seems like you're stuck with two rigid options. Option one means hitting your macros every day for the rest of your life, and option two means eating that chicken breast and broccoli at exactly 3 pm for the rest of your life. For most people, neither is sustainable long term and both share the weakness of promoting a black and white mindset; you're either on your meal plan or *off* it, or you either hit your macros or you didn't (see "Magic Macros" in the Mindset and Materials section). Fortunately, this doesn't have to be the case.

The solution for those who aren't dieting or for non-competitors looking for long-term plans is to seek non-binary approaches. Meaning, it's not either a zero or a one, black or white, or on or off. Rather, you have a continuum which you can follow and options to use based on the demands of the given situation. Early on in this book I explained the importance of developing this mindset, now I am going to give you a system to apply it.

Defining The Tiers

So even though we might aim to be as precise as possible, life happens and it's usually not practical to be exact all the time. With a binary approach, a slight deviation results in "falling off the diet", but a flexible approach allows for other outcomes. For example, when a target is not perfectly hit, this simply means that you broaden the target by dropping to another level on the precision scale and carry on.

To build upon this notion, here are the basics of our 3 nutritional tracking tiers:

- Good: You hit your target calorie goal within a certain + or - range. Typically I recommend using + or -100 kcals for your calorie goal.
- Better: You hit your protein within a certain + or - range and also hit a target calorie goal within a certain + or - range. Use the same ranges for protein above, and use + or -100 kcals for your calorie goal.
- Best: You hit your macros within a certain + or - range. For contest prep I would recommend + or -5 g, for other types of cuts I would recommend + or -10 g, and for the offseason I'd recommend + or -20 g on carbs and protein and + or -10 g on fat.

This approach is best used when tracking macros is not a major stress for you. Most of the time, you try to do "Best", but remind yourself that both "Better" and "Good" are totally acceptable options that can occur on a regular basis. This approach allows for social events, going out to eat, holidays, and drinking alcoholic beverages. It's also handy for times when you accidentally went over on one macronutrient target, or when hitting your protein target would put you over your calorie goals for the day.

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Borrowing

Normally we are accustomed to sticking to our targets in a 24-hour period. This might be appropriate during contest preparation where most days are low in calories and at a certain point you are walking the "tight rope". However, for casual diets or the offseason, when you have ample glycogen and body fat stores and plenty of calories to play with, you can be more flexible. Take the 24-hour period off the pedestal and all of a sudden you have more options.

With the borrowing approach, you can take up to 20% of the macros/calories on any one day, and give it to another day. That way, if a planned event is coming up or if you simply are very hungry on one day or not hungry at all on another day, you can shift your calories around. A 20% reduction on any isolated day in a non-dieted and depleted body is going to have very little impact on anything of import, and it allows one to mix and match to account of the occurrences of real life.

Another way to institute borrowing is to set up an excel sheet that calculates a 7-day average of your calories for the week. You can simply have the goal of hitting the target calories on average by the end of the week. When doing so, by the end of the 7-day period, if your daily goal was 2500 kcals for example, so long as you averaged that over the 7-day period, you've met your goal for the week. This way, you can have a day at 3000 calories, a day at 2000 calories, a day at 2700, a day at 2300 and then the remaining three days at 2500 calories and that's absolutely fine, rather than having to rigidly consume 2500 calories every day.

Relearning How To Listen To Your Body

If you are a competitive bodybuilder, there is a certain point during contest prep where you have to stop listening to your body. Your hunger signals become constant, and your satiety signals disappear. In these cases, your plan is the only tool you have to ensure you don't overeat.

However, when you are not in a contest prep phase, there should come a time to let go a bit. If you spent months weighing and tracking your food and bodyweight, looking at nutritional labels, learning where calories come from, and changing your eating habits to reach nutritional targets, you are better equipped than you once were. But, at the same time you've gotten so used to following a relatively rigid plan that doing so can become second nature and can even replace what normal humans use to regulate their energy intake: hunger and satiety.

Ironically, after you spend years "going by the numbers" and modifying your body composition using the quantitative approaches outlined in this guide, I believe it is actually very important to then learn how to once again listen to your body. The goal of integrating the awareness of your hunger and satiety levels with your newfound nutritional knowledge and experience is to eventually develop an approach that requires minimal day to day effort. The first step in checking if you are ready for this would

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be to stop following your nutritional plan for a day. Don't try to hit your targets, just simply eat, but write down the foods you consume and rough portion sizes (but not the macros or calories).

At the end of the day, without looking at what you wrote down, estimate in your mind your macros and calories for the day. Then use the notes you took on what you actually ate to determine your true intake and compare the two. If your recollection and estimation was pretty close to what you actually consumed, and you didn't go hog wild and over eat, or drastically under eat, you are most-likely ready for this approach of listening to your hunger signals a bit more.

Tracking Your Diet With Just Habits And Bodyweight

First of all, let me make it clear that the following strategy is probably most appropriate in the offseason when hunger and satiety signals are operating. That being said, once you've established that you can accurately gauge your calorie and macronutrient intakes, then you need to see how close you were to where you want your numbers to be. If you are accurate with tracking macros and calories but your protein is habitually low, or if you are accurate but you aren't eating enough calories, or if you are accurate but you never eat fruits and vegetables, you have just identified the things you need to mindfully change in order to reach your goals with this approach.

This identification system, along with bodyweight monitoring, can help you adhere to a plan that is far less constricting than what most people would consider a typical muscle and strength type of diet. If you'd like to maintain a consistent trajectory towards your long-term success without the pressures of tracking all of your dietary numbers, I suggest a simpler approach as detailed below.

The following bullet points below offer some guidance on how to monitor habits and bodyweight for nutritional success. Once again, this approach should only be taken after you have a lot of experience with tracking, measuring, weighing and manipulating your body composition. If you have this experience and have established that you can be accurate on "auto pilot", then you can use this approach and the following advice:

- If your goal is to slowly gain weight and you slowly lose or maintain weight when following your habitual eating patterns, strive to be a little full at most meals or during most days of the week. Check your weigh-ins to make sure you are successful in achieving slow weight gain and also not overdoing it and gaining weight too quickly. This approach is not appropriate for extended cuts to get very lean as you will very quickly not be able to trust your hunger and satiety signals.
- Likewise, if your goal is to slowly gain weight or maintain weight and you gain weight too quickly, try to consciously decrease portion sizes and stop just when satisfied at meals. Also try eating slower to allow time for your satiety to catch up. Once again, double check this strategy with your scale.

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- Don't stress your carbohydrate and fat intake. Unless you are really low on one or the other (which typically takes conscious avoidance and effort), just eat. If you consumed 20% of your calories from fat on one day and 40% on another day, it likely doesn't matter one bit in the grand scheme of things when you aren't dieting. So, just focus on calories and protein.
- If you are an offseason strength athlete, you want to make sure you aren't too low in protein. While there is good reason to eat high protein diets while dieting [13], you are just fine in the range of 0.8-1.0 g/lb when you aren't lean and calorically restricted [14]. If you find you habitually undershoot your protein, try adding a protein shake or two per day to get up to this range. If you are over this protein range, that's fine there's no harm in it so long as you aren't eating a very low fat or very low carbohydrate intake because of it.
- Remember the micronutrient level and the recommendation to try to consume at least one serving of fruits and vegetables for every 1000 kcals you eat. If you find you don't do this habitually, buy fruit and leave it easily accessible and make a point to pack a couple pieces of fruit to take to work or school. For vegetables, try to have a salad each day and that will most likely cover your servings of vegetables. If you can hit these targets, you're most certainly going to be consuming enough fiber as well.

These systems above can be used in isolation, or in conjunction. For example, you can have a day where you are eating targets just based on calories and protein (the "Better" option from the three tiers), and also borrow 20% of your calories from that day to give to another day of the week. Likewise, you can follow the habit development game plan, and then intermittently go back to tracking using the 3 tiered system and borrowing to ensure you were being accurate and staying on track.

None of these systems are mutually exclusive, and when used together they provide you more options. If you can learn these principles and slowly implement them into your lifestyle, you will most likely have a much easier time with your athletic endeavors in the long run and will be able to live a much more normal existence than if you perfectly hit targets every single day.

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NUTRITIONAL TRACKING TIERS *In descending order based on precision*	
BEST	**Hit your macros within a certain + or - range.** Contest Prep: + or - 5 g of each macronutrient Fat Loss Phase: + or - 10 g of each macronutrient Off-Season: + or - 10 g for fat and + or - 20 g for protein/carbohydrate
BETTER	**Hit your protein AND calories within a certain + or - range.** Protein: Same as "BEST" tier Calories: + or - 100 kcal
GOOD	**Hit your target calorie goal within a certain + or - range.** Primarily used in off-season: + or - 100 kcal
Habit-Based Tracking	**Aim for consistent implementation of habits after establishing accuracy.** If you constantly undershoot protein, over-eat dietary fat, miss out on fruits and veggies, or fail to execute any other beneficial behavior, make it a point to monitor yourself around that one thing until you can achieve it consistently. Once mastered, find another behavior to alter in your favor if necessary. Continue until healthy, helpful habits are established in agreement with your goals. Best combined with tracking bodyweight (using 7-day averages).

Eating Out

So now that you've got all the different forms of tracking under your belt, it's important that you are able to implement them outside of your own kitchen when out with friends and family. But before we dive into food choice strategies, let's address all of you in-season competitive bodybuilders.

Simply put, during contest prep, eating out is something you want to avoid. The times I suggest that you do eat a meal out during a competition diet is when you have something that is very important in your life.

For example, in 2011 I dieted from January to August, and I ate out 3 times: for my anniversary with my wife, for Mother's Day, and for my mother's birthday. Each time I went out, I got a very simple meal. I had double steamed broccoli and a lean sirloin steak with no butter. I didn't weigh the food, I just estimated the calories and macros. But, I knew that even if the cooks didn't ensure that I got exactly a 7oz sirloin steak and exactly 2 cups of broccoli, that at most my estimations would be off by maybe a couple grams of carbohydrate and maybe 6 or 7 grams of protein and fat each. So the consequences from estimating that meal's macros incorrectly, if any at all, were very low.

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However, if I had ordered pasta, that could have been off by massive amounts. It's so easy for a chef to put an extra 2 tablespoons of oil above and beyond what the nutrition information for the restaurant lists. Adding that oil alone means you could have an extra 250 calories that you didn't plan for and most likely wouldn't even notice simply by taste. Even if I could have fit the macros of the pasta dish into my diet for that day, I would have been making a big gamble assuming that the macros I estimated or that were listed on the restaurant's menu were accurate.

So just remember that chefs and cooks are not going to weigh your food out for you. They don't care about your macros, they care only about giving you a tasty meal. So when you eat out, be aware that the nutrition information won't be accurate, and your best guesses can be way off if you order foods that are likely to have a large margin of error (due to being energy dense). However, there are ways you can mitigate the risk by making smart choices. On the whole, if you are a competitive physique athlete, I would limit eating out to once per month at most.

Now remember, I am talking about the specific case of contest preparation for a competitive bodybuilder. I am not saying that those of you who are dieting for non-competitive purposes should strive for this level of control. In fact I don't think you should, it would likely hamper adherence. But in the end, competitive bodybuilding is an extreme sport, and thus even when approached in a flexible manner, it takes a relatively extreme level of commitment to achieve success on stage.

If you're not in contest prep, if you're just doing a cut, eating out 1 to 2 times per week is fine, so long as you make conservative estimations of the caloric and macronutrient contents of your food and choose low-calorie options (like the steak and broccoli versus the pasta). Doing so will ensure that you are accurate while still participating in important social aspects of your life.

If you're in the off-season, eating out 1 to 2 times a week is also fine (or even more often). Track the meal with your best estimations, but don't worry too much about the macronutrient breakdown. You have a larger calorie buffer, so you may be able to work it into your target macros (if you have them when not dieting) by the end of the day. And even if you're not able to hit your macros, just fall back to "Better" or "Good". Likewise, if you accidentally go over your calorie range and can't fall back to the "Good" tier, use the 20% borrowing rule and pull from the next day's calories (up to 20%).

Remember that if you're eating 3 to 5 meals a day you're going to have an average of 28 meals per week. If 1 to 2 of them is only 50 to 75% accurate, it's not going to be a big deal in the grand scheme of things in the offseason. So long as you are getting in enough protein, and gaining weight at an appropriate rate, you have covered the most important offseason nutrition variables.

Alcohol

Simply put, the key with alcohol is moderation. No matter how popular it is in college

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or how common it may be in certain cultures, binge drinking until you are completely wasted or throwing up is not a healthy habit. It's also not normal or conducive to an active healthy lifestyle.

That said, what does moderation mean? Well in my opinion, it means drinking to the point where you don't feel it the next day. Or if you did feel it slightly in the morning, it certainly wouldn't hamper your performance. In addition to that guideline, I'd also say that it would need to fit within your dietary goals as well.

Although we did not mention alcohol in our macronutrient section in Level 2, it definitely has calories. To be precise, it actually contains 7 calories per gram, and most of the time it's combined with carbohydrates. Wine is made from fruit and beer has hops, wheat and barley. All of which are carbohydrates, which also have calories.

Now that said, we don't have a fourth macro that we follow. So when you decide to drink you have automatically fallen to the "Better" or "Good" tier. Since you don't have an alcohol target, consuming alcohol means you automatically revert to calories and protein, or just calories.

So let's say you had a couple of beers which ends up being 400 calories total. Your macros are normally 200 protein, 300 carbs, and 70 grams of fat. In terms of calories, this amounts to 2630. By consuming alcohol, you now are simply focused on either hitting calories and protein (tier 2 "Better") or just calories (tier 3 "Good"). After the alcohol consumption, you would have 2230 remaining to spread between protein, carbs, and fat for the day (finishing by hitting your protein target and calories within a + or - 100 kcal range). This way you've still met your number one goal of energy balance and you've ensured adequate protein. This is also a moderate amount that wouldn't affect your training the next day, so you're all good.

Just like eating out, I would recommend drinking only once or twice a week, in moderation. If pressured to define "moderation" further, I would recommend consuming no more than 15% of your total daily allotted calories from alcohol. For most people that amount is one, two, or maybe three drinks if you're a big guy in the offseason with a lot of calories to play with.

So as you can see, this alcohol recommendation, along with the entirety of this section was very broad and theory-based as opposed to the hard-set prescriptions from the actual Pyramid Levels. Although there is not an absolute right or wrong here, some general advice would be to err on the side of being overly precise as opposed to being overly flexible at the start of your journey when you are less accurate at tracking.

Then, over time as you get more aware of the calorie and macronutrient contents of foods you will be able to be more flexible while still having appropriate accuracy. Eventually, you will find your sweet spot that works best for you at different stages dependent on your immediate and long-terms goals as an athlete. Just always remember that precision and tracking will need to follow a wider road at times as opposed to a tight rope in order to maintain a sustainable lifestyle.

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Social Environment

The entire purpose of this text was to give an evidence-based approach to some often-debated topics related to some not-so-common ways of living. Because you have chosen to embark on a different path than a lot of the rest of the world, it is quite possible that you will receive some resistance from various people in your life when they witness or are affected by your personal choices and habits.

Given that, I think it's important for me to leave some of my anecdotal recommendations for physique and strength athletes to better exist in society. I'm not simply talking about how you can make it through your day, because if you were able to purchase this guide you are probably doing alright in terms of basic survival. Rather, what I want to cover is how you can progress towards your athletic goals while maintaining an emotionally stable existence, and stay happy without alienating people in your life who deserve the best parts of you.

Support

From birth, we are obviously unable to do things on our own. Throughout the first few years of our lives, this continues on for survival. But beyond that, no matter how much people want to *think* they can do everything on their own, it is simply not the case. Humans have evolved to be social creatures, and it is imperative that we maintain relationships with other individuals in order to lead fulfilled lives.

This is not just about personal happiness, but also about your success as an athlete. Yes, I will argue that having a stable and supportive social environment aids your ability to increase your muscle and strength to some degree. The optimal amount of social engagement might be more or less for you depending on how introverted or extroverted you are, but human connection and support is necessary to some extent for all of us and it is imperative that we strive to balance this aspect of our lives and integrate it with our goals. Scientifically, there is a great deal of research that shows the beneficial effects of a positive support network on behavior change for nutrition and exercise [15-19]. It is this reason that we emphasize *Team* 3DMJ, not just 3DMJ in our coaching model.

Framily

This word, "framily", is a catch-all phrase for your friends and family; your wife, your dad, your best friend, your coach, your colleagues, all the people who make up those you care about and whom care about you and your life.

They can only support you if they understand what you're doing, what you're going through, how you experience it, and what it means to you. I've run into a lot of people who start weighing their foods, start losing weight, and possibly even start dieting to get extremely lean for a competition without having some basic initial dialogue with their framily unit. Neglecting to do so can generally cause athletes to lose the support of their loved ones just a few shorts weeks into the dieting phase.

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And this is not because their framily doesn't care about them, it's because their framily doesn't understand. So, I would really suggest that you enlist the aid of the people you care about and who care about you. Obtain a support network if you don't already have one and give them the communication they deserve, so they know what journey you are embarking on.

A support network can be online, at a local fitness club, at Weight Watchers, or anywhere really; there's a lot of ways to do this. But get your support, find your framily, and explain to them what you're doing in a calm, collected, informative manner right from the beginning of your journey.

Communication

Once you've decided to have this conversation, you want to include not only what you are doing, but why you are doing it and why it is important to you. Your framily doesn't have to be a fan of bodybuilding or to "get" what you are doing necessarily. However, they do need to understand how you feel about it, why, and how important it is to you. I have a lot of framily members who flat out don't like bodybuilding. That's okay! What is important is that they love and respect me, and they show me support, even if they don't necessarily "get" the sport I compete in. Also, be explicit with your desires. For example, if you want the support of your loved ones you could say "I would really appreciate your support in this. It's going to be difficult for these reasons, at these times, and it would mean a lot to me just knowing that you have my back."

Don't make assumptions about what they do or don't know about what you're doing or why you're doing it, and likewise don't make assumptions about why they're acting the way they're acting. Take responsibility for your choices and have dialogues with your framily on occasion throughout the process. Adult communication means stating what you need, informing the people you care about as to why it matters to you, and then asking for feedback on how you are holding up your end of the relationship as well.

It is extremely important to understand that asking for help is not weak. Anyone who is smart and who is honest with themselves shouldn't be afraid of looking weak. Showing your vulnerabilities takes true courage and puts you ahead of most people who don't show or admit to any vulnerability, and therefore can't meaningfully connect with others. So, do yourself a favor as an athlete and communicate effectively, get the support you need, and interact with your loved ones responsibly in an emotionally mature way.

Also, remember that you are choosing to do this. If you are dieting, or competing, that is your choice, and it is not reasonable to expect the world to change for you. Being emotionally healthy and being clear with your needs and expectations is not the same as acting as though you are entitled to different treatment because you voluntarily chose to get on a bodybuilding stage.

One of the goals I try to encourage my athletes to pursue, is to try to leave as small of a social impact on those around them due to the process of dieting. Meaning, they

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try to live their lives during prep in a way that minimally affects their loved ones. This might mean still going out with the boys or girls on a weekend, but instead of drinking, offering to be the designated driver and ordering diet soda but still laughing it up and connecting with your friends. Or, instead of just eating your own pre-prepared meal at home and never taking your significant other out to eat, maybe you instead cook a meal and have a date night in and watch a movie.

Perhaps, even though you don't go out on dinner dates as much, you might go on movie dates. Or maybe, you do occasionally go on dinner dates during your diet, but you make the right choices so it doesn't get in the way of your nutritional targets. Proverbially (but probably not literally unfortunately), the goal is to have your cake and eat it too.

Helping Others

Although your framily might not always know that it is helpful to ask questions and have regular dialogue with you about your health and fitness goals (until you train them to, of course), there will also be a select group of people on the flip side who cannot help but ask for advice and input from you. These individuals can typically be found at the gym where you train or amongst coworkers. I'm talking about the people in your daily life who notice your progress and want similar results for themselves.

Some people do a great job interacting with these people in an empathetic, intelligent, and open-minded way. Others, not so much, and often alienate, shame, judge or even give out inappropriate advice to those they interact with. If you are someone who wants to be a leader in the fitness field, it is imperative that you understand that what you say can carry a lot of weight in the eyes of those who are seeking answers.

Here are some thoughts and strategies to help you carry on your training, your reputation, and your integrity without alienating or upsetting those around you.

Challenging Convention

Some of the things I've said in this book challenge the conventional wisdom of the old school. So, when people ask you for help, don't be surprised when sometimes they express shock that you aren't doing things that they think are required for success (such as banning certain foods or food groups, having rigid meal timing or frequency, etc). So, when you respond to questions do it in a way that you are not intentionally bashing tradition just for the sake of feeling smart or clever.

An example of handling this situation poorly are the common interactions you see when the topic of "if it fits your macros" vs. "clean and dirty" dietary approaches come up. These conversations often lead to shame and bashing rather than education, which simply creates more deeply entrenched divisions. Both groups are trying to get healthy and be in better shape, but by developing two hard lined camps that throw stones at each other, neither learns anything from the other. What many of them don't know (or choose not to acknowledge) is that there are flawed philosophies on both sides.

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You don't have to tell someone else they're stupid to answer a question. You don't need to "myth bust" and attack someone's idols to get across new concepts, you can just simply tell them what to do and the logical reason why to do it. Don't approach conversations with a condescending mindset or tone of voice. When people ask you what you're doing or say, "Hey I thought (insert inaccurate claim here)", approach them without a shaming or condescending demeanor. If you do bash what they were doing in your explanation, you are more likely to put them on the defensive, embarrass them, and you are more likely to entrench them against the message you want them to learn.

Just try to explain to them what you're doing and why. And if they want more help, direct them to some of the resources that you learned from, and/or share more of what you do at your leisure. What you don't want to do is give out unsolicited advice or bash the opinions of anyone else who tells them different information than you do.

Unsolicited Advice

Let people come to you rather than approach them unsolicited, because that indicates they are ready to learn and try something different than what they've been doing. If you're sitting on the bench press waiting between sets, odds are if you even eavesdrop just a little, you're going to hear things that will make you want to face palm. Fight the urge of interrupting someone else that is either giving advice or sharing what they are doing by giving unsolicited advice.

Understand that the person you hear saying these not-so-accurate, but very common philosophies, is not intentionally doing anything bad; he or she is just trying to help somebody reach the same goals we all want to reach. It's also worth pointing out that what the person receiving the "bad advice" was doing before (which may even be nothing) is possibly worse than even the incorrect or unnecessary advice they are receiving. So the simple fact that they're getting involved, asking questions and taking a step forward is a good thing.

Please remember that the "evidence based" fitness professionals and the so called "bros" are on the same team. It's just different mindsets and different approaches. If you don't corner yourself and put a label on what you're doing and what they are doing, you can actually learn from one another and the fitness community would actually see progress in more areas.

In short, be open-minded, compassionate and don't give unsolicited advice. Also, you don't need to make someone else feel stupid or wrong to give them advice when it is solicited. Just share your knowledge in a non-emotional, non-attacking, non-confrontational way when asked for it and you're more likely to actually effect positive changes in that person's mindset.

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RESOURCES

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I'd like to close out this guide with a list of resources for your further learning and application of the material discussed in The Muscle and Strength Pyramids. Please take the time to examine the following information, links, and people at your leisure for additional reference and in depth discussions on training and nutritional topics involving physique and strength athletes.

Click the underlined titles below to visit the appropriate websites.

NUTRITION CALCULATION AID

Here is the spreadsheet calculator to help you with the nutritional calculations in the book. The password for the page is "nutrition-aid".

Contributors To The Pyramids

Here are the links to access other resources available from myself, my coaching team, and my co-authors, Andrea and Andy. Without these ideas, videos and people, the creation of this pair of books would not have been possible.

NUTRITION PYRAMID VIDEO SERIES

As discussed in the preface of this book, the Muscle and Strength Nutrition Pyramid was originally presented as a video series that I created for the 3D Muscle Journey YouTube channel back in 2013. Many of the ideas are similar, and watching those videos may be useful for some people's absorption of the material. However, be weary in that some of the information may be outdated, which is why I have created this updated guide with current recommendations in accordance with more recent scientific findings.

TRAINING PYRAMID VIDEO SERIES

In the same fashion as the Nutrition Pyramid above, the Muscle and Strength Training Pyramid video series was published on the 3D Muscle Journey YouTube channel in May of 2015. As you've probably guessed, it was also a whiteboard presentation to give people an order of operations in regards to deciphering what matters most for their physique and strength building goals.

3D MUSCLE JOURNEY WEBSITE

This site is the home base for all 5 of the coaches from Team 3D Muscle Journey. Here you can find blog articles, videos, and information on coaching from Jeff Alberts, Brad Loomis, Alberto Nunez, Andrea Valdez, and yours truly. We offer Contest Prep Coaching for physique athletes, Meet Prep Coaching for powerlifters, and one-on-on Skype-Based Programming sessions for non-competitors or off-season athletes.

3D MUSCLE JOURNEY YOUTUBE CHANNEL

The bulk of our online content as a team is housed in video format over on the 3DMJ YouTube channel. Here you can find an abundance of free instructional videos and

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video diary series from many of our coaches on their very own athletic journeys.

ERIC'S RESEARCHGATE PROFILE

ResearchGate call themselves "The professional network for scientists and researchers", so that's what I use it for. Here you can find all of my publications, co-authors, abstracts, and contributions as an investigator in the field of nutritional sciences and human performance. When I am legally allowed, I also post up the full texts of the articles I have published. When I'm not allowed, you'll at least be able to read the abstracts and send me a personal message in regard to any of the publications.

ANDREA'S WEBSITE: ANDREAVALDEZ.NET

As a fellow 3DMJ coach and longtime athlete, Andrea's personal website offers a look into what it takes to be a competitor. Along with the articles about training, nutrition, and mindset from her personal perspective, the website offers some of the best contest prep eBooks you can possibly find online.

ANDY'S WEBSITE: RIPPEDBODY.JP

The content of Andy's site is delivered as an all-encompassing guide on how to adopt a straightforward, no-nonsense approach to your nutrition and training. You can find information about Andy's online coaching services, and how you can get a copy of his book dedicated to making diet adjustments for your goals, *The Diet Coaching & Adjustments Manual*, which is an excellent companion to *The Muscle and Strength Nutritional Pyramid*, and a book I've contributed to as well.

Tools & Databases

EXAMINE.COM

This site proudly claims itself as "your unbiased source of nutrition information", and I'd have to agree 100%. Their database of dietary supplement information is unparalleled online, and the absolute best part about the depth of their research is the way it is presented in the Human Effect Matrix, a tool for informing readers how strong the evidence is to support the proposed outcomes of these supplements.

MYFITNESSPAL.COM

Of all the food calculating databases, I'd say that MyFitnessPal is the most accessible and the most complete one on the market. Although there are options to pay for upgrades, the standard free web and mobile app does everything you'd need it to. Track your calories and macronutrients manually at home, using a barcode scanner, or by using their large database of common restaurants and eateries as supplied within the app itself.

EATTHISMUCH.COM

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Just to reiterate from the Nutrition Pyramid... No, I do not think that meal plans are a way of life or that they should be followed forever and ever. That said, if you are just starting out on your journey to learning how to manipulate your foods and would like some example meals plans as training wheels, I think this is the best website for that.

Companies, Courses, & Publications

JISSN.COM

This is the home page for the Journal of International Society of Sports Nutrition. This peer-reviewed publication is unique in that it is entirely open access. That means you can read the full texts of any the studies published in this journal. The articles published in the JISSN can give insight into the acute and chronic effects of sports nutrition and supplementation strategies on body composition, physical performance and metabolism.

CITADELNUTRITION.COM

I wanted to include Citadel Nutrition's website because it is one of the very few companies that I have found to give such transparency and attention to detail when it comes to the quality and integrity of their products. Their products are evidence based, and they also go out of their way to get the input from evidence based professionals in the field to ensure they are doing right by the customers. To give you an idea of their integrity, they are the only supplement company that I have written articles for. No, I don't think supplements are necessary, but if you were going to purchase some, take a look at Citadel's site first.

STRENGTHANDCONDITIONINGRESEARCH.COM

S&C Research is an encyclopedia of strength and conditioning methods for improving muscular strength, size and athletic performance. This is a monthly subscription to receive the latest research, trends, and summaries of unmissable information as curated by two big names in the online fitness industry, Chris Beardsley and Bret Contreras.

THESBSACADEMY.COM

The Shredded By Science Academy is labeled as "A year-long evidence-based academy for personal trainers, brought to you by the best educators in the fitness industry" and I couldn't agree more. The man behind SBS is Luke Johnson, a leader in the UK in terms of continuing education for personal trainers. As one of the main contributors to this course along with Dr. Mike Zourdos, I feel like this is a very valuable investment for anyone looking to enter the physique or strength coaching industries.

ALANARAGON.COM

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Alan Aragon's monthly Research Review is perhaps the best $10 you'll spend each month. Alan distributes very useful, actionable insight from his own experience, as well as from the best researchers, scientists, and industry experts in their respective fields. This is a must-read for anyone interested in science-based approaches to fitness, physique, and strength.

Other Awesome Websites Worth Checking Out

BODYRECOMPOSITION.COM

The home of Lyle McDonald. This site has been around for donkey's years now and still remains one of the most thorough non-textbook sources of information on nutrition and training. Lyle's books are all worth buying, as you'll learn something new from each one. He understands all the science, and is able to find unique ways of applying it to help people reach their goals. I would also recommend staying on the lookout for Lyle's two new upcoming books, one on nutrition and training for women, and the other that he plans to make his encyclopedic opus on everything fitness related. I was fortunate enough to contribute a small amount to both of these texts and I can say they are nothing short of epic. The forums on bodyrecomposition.com are also legendary, and if you go back ten years you'll see many well-known names now who were nobody's back then - including yours truly. :)

STRENGTHEORY.COM

This website, primarily authored by former IPF world record-holding powerlifter Greg Nuckols, has become one of the best sources of original content on strength training on the web in the last couple of years and is growing rapidly. Check it out.

LOOKGREATNAKED.COM

Researcher Dr. Brad Schonfeld's blog. A good opportunity to see an exercise scientist with a lot of experience in the trenches as a trainer discussing their latest research and the limitations in a transparent and authentic manner. His applied research is well-presented for those who find some scientific literature daunting or difficult to understand.

REACTIVETRAININGSYSTEMS.COM

Founded by World Champion IPF raw powerlifter Mike Tuchscherer, Reactive Training Systems is a wealth of knowledge for anyone interested in gaining strength. Not only are they publishing content at the forefront of autoregulation in strength training, they also provide custom coaching services and a variety of products to further your programming education.

THESTRENGTHATHLETE.COM

The Strength Athlete has quickly become one of the most successful coaching services

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for raw drug-free powerlifters in their industry. Founded by one of my personal 3DMJ coaching clients, elite lifter Bryce Lewis, this site offers weekly coaching, online consultations, and an abundance of free articles and programs for lifters of all levels. Bryce is a master when it comes to creating resources for the drug-free raw powerlifting community. He has a unique combination of an analytical yet creative mind, artistic talent, and has a lot of experience as a coach; currently for powerlifting and previously for volleyball. He was the artistic designer for the pyramid images that appear in both books, and the excel-wizard who created our webpage where purchasers of The Muscle and Strength Nutrition Pyramid can setup their nutrition plan.

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Final Words

Final Word From Eric Helms

First, let me just say thank you for reading. The fact that you read through this entire book gives me a profound sense of accomplishment. While it may seem like just an eBook to some, this book represents over a decade of my personal experience. The hours, days, weeks, months and years spent reading scientific articles and textbooks, performing experiments, writing and reviewing published papers, teaching personal trainers, teaching undergraduate and graduate exercise science and nutrition students, presenting at conferences, debating and discussing topics on and offline, helping athletes develop from novices, to intermediates, to elite competitors, to world champions, and the years of putting the theory to practice as an athlete myself, has culminated in what you just read. While it is important to me that you benefit from this information whether you are an athlete, a coach or a recreational trainee, I also want you to know that just by reading this book you've taken the time to step into my experience. For that, I want to thank you as that act is an incredible sign of respect that I don't take lightly.

Also, I hope that the big picture concepts in this book have shifted your perspective for the better. I hope now that you are able to differentiate between what is most important, and what might not matter at all. I also hope that you can shift away from black and white thinking, and instead understand the aspects of context and individuality that will help you surpass your goals and reach new goals in a much more time efficient manner.

Finally, I want to take the time to say thank you to all the people who have helped me develop into the person I am today and influenced my development as a professional coach, educator and researcher. You know who you are, and in fact I've probably told you as much in person or online. But, I want you to see this book in many ways as your own, because I truly could not have done this without you. To Andrea and Andy, a huge thanks for providing the structure, organization, and motivation to put this together. This book is infinitely better because of the two of you and you both have a drive and a passion that is rare in this world, and I am and grateful that we are on the same team!

To the reader, I'm sure you will be hungry for more and I encourage you to check out the resources section of this book that will help you flesh out your understanding of the concepts you learned here, and that will also complement this information. Also, be sure to keep an eye on The Muscle and Strength Pyramids website as well, as we have resources available to those who purchased the books. Additionally, our website will be updated with useful information, links, and blog posts and you can also contact us through it if you have any pressing questions.

Once again, thank you and good luck!

Sincerely,
Eric

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Final Word From Andy Morgan

I feel very fortunate to have been asked by Eric to work on this project. It was probably the happiest moment of my career.

There are certainly others who are far more qualified for the job, and when you consider that he could have asked nearly anyone in the field to co-author a book with him, it would be fair for you to wonder why he chose me.

I put it down to one part luck, and two parts karma.

The reason that you have this book in your hands now, rather than having to wait another year for it, is thanks to a superstar called Kengo Yao who works with me on our Japanese sister site, athletebody.jp. In early 2015, we approached Eric to ask if we could make a short eBook version of his YouTube "Training Pyramid" series in Japanese to give away free to our audience. Eric graciously gave us the go-ahead, so Ken spent a solid 5 weeks working on it night and day, and it was a hit. The quality of the structure and content so impressed Eric that he asked me to help him put together an English version, which turned into this book project.

The Japanese site wouldn't still have been around at this time had I only been concerned with making a profit, as it was run at a loss until the end of 2015, its fourth year of operation.

It would be easy to look at our site now - the audience figures, contributors, and respect we garner - and forget about all the hard work that went into it. Nobody gave a damn for the first two years, but what kept me going is threefold:

- Firstly, Ken was willing to accept a salary lower than what he was truly deserving of, but what I could afford.
- Secondly, I could see it becoming truly industry-wide impactful one day if we persisted with the fight against the broscience here.
- Finally, and perhaps the most important to me personally, I had the support and permission of everyone I look up to in this industry to translate their articles. This developed into friendships which I didn't quite feel deserving of, and there was no way I was prepared to turn up to conferences each year in the US having let them down.

The nutrition book's content is 99% Andrea and Eric's hard work, I can't take credit aside from some very small suggestions. To say that I wrote the training book with Eric would also vastly overstate my contribution. Ken's original layout formed the base from which I created the first badly written draft, which Eric was then forced to almost completely rewrite. My main contribution has been to pull more and more information out of Eric at every revision, and to get him to write in a way that is simple enough for even a dummy like me to understand.

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I've tried to deliver valuable contributions by constantly adding, re-editing, and suggesting new content. I also compiled everything in PDF format and put together the website, which I look forward to building up over the coming year(s).

Overall, this project has been a humbling experience. I've been coaching people for a few years now, and I thought I knew the topic well enough, but this collaboration has turned out to be a real blessing for my own education.

I'm proud of what we've put together as a team. From the heart, I'd like to thank you for picking up these books, and I wish you all the best with your fitness journey. If you ever feel the need to get in touch, I'm always available over on my English language site, RippedBody.jp.
Andy

Final Word From Andrea Valdez

Now that you have made your way through this guide, I'm sure that you are completely sold on the fact that Eric Helms is simply brilliant for a number of reasons. The only thing I'd like to take credit for is recognizing this information early enough to do something about it.

Back in 2011, when I first discovered the wonders of YouTube, Eric and the other coaches of Team 3D Muscle Journey began changing my life from afar with their unconventional outlook on training and nutrition. It was because of these four gentlemen that I was able to come out of my own post-contest depression and get my normal existence back.

The next year I began sharing what I had learned on my own website and was accepted as one of their athletes for my 2013 contest season. I moved across the country to be around the 3DMJ buzz of coaches, teammates, training meet-ups and competitions for the following two years. By 2015, I had worked my way into becoming the fifth and only non-original coach to be added to Team 3D Muscle Journey and I could not be more humbled to call them my family.

I don't tell you this to brag (although I really am quite proud of the company I keep), but more so to let you readers know how much it truly means to me that you have taken the time to read this.

This is not a simple project that took a few months to put together, it actually originated from years of writing, failing, succeeding, competing, suffering, investing, learning, and evolving. Yes, I am now a part of one of the best natural bodybuilding and strength teams in the world, but the basis of all my education began with those YouTube videos so long ago.

And one video series in particular (Eric's Nutrition Pyramid, obviously) was the catalyst that pulled me out of many psychological and physiological issues that had arisen due to my chronic under-eating and lack of dietary prioritization...And *that* situation is a prime example of why it was so important for me to help bring these books into the

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world.

I know the value of this information you have just read, and I consider it one of my life's biggest achievements to have helped spread these ideas to those who need to hear them. This is not about telling people how to "be optimal" with their training and nutrition, but more about giving them the tools to navigate through their frustrations, failures, and mental hang-ups in order to proceed toward their goals with minimal negative outcomes. This mission is a damn important one and I am happy to share it with Eric and Andy.

I can only hope that you have enjoyed our work as much as we have enjoyed putting it together for you. Eric's incredible breadth of knowledge mixed with Andy's prowess with all things internet business enabled me to transcribe and edit my little fingers to the bone for these little gems to become what they are. Our combined efforts have turned a few white board cell phone videos into the most comprehensive manuals in our industry to date.

But all that said, I think it's high time I acknowledged you.

For those of you reading this, those who have invested time in their education, I'd like to thank you for being the type of person whose curiosity will continue to further this field of study for decades to come. It's people like you, those who truly have a passion for science-based learning with an open mind, that will also be the role models, leaders, researchers, and spokespersons for people who truly need it in the future. It's quite a large responsibility, but after reading these guides, I think you can handle it :)

With tons of love and appreciation,
Andrea

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